

Technical Validation Supplement

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the analysis note

Draft 1.5, release 2026-09-06 · document revision 2026-09-07

release 2026-09-06 · 51 differential extractions + 6 one-bin totals · blind

Ishanee Pophale, Jarek Nowak

Lancaster University

Abstract

This supplement holds the validation and diagnostic material for the $\text{CC}\pi^\pm$ NuMI cross-section measurement: the studies that establish what the measurement can and cannot resolve, the estimators that were tried and rejected, and the closure tests that qualify the extraction. It is separated from the analysis note so that the note describes one current analysis state, while the record of how that state was arrived at remains available and citable.

The two documents were generated from a common source and are checked for numerical consistency where current results are duplicated (`report/check_tables.py`, `xsec_analyzer/check_staleness.py`, `check_staleness.py`); that is a check, not a guarantee, and the change log records what each round of review found. Superseded material *is* retained here where it is evidence — pre-fix figures, withdrawn model comparisons, rejected binnings — but only as labelled historical diagnostic record, and none of it may be quoted as a current result. The corrections that superseded it are recorded separately in the change log, with before-and-after numbers.

The central results of this supplement are negative ones, and they set the scope of the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ c and carries little information about the initial momentum above roughly 0.4 GeV/ c . Every alternative estimator tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — fails to recover it, and a truth-level golden selection performs no better than the classifier, which places the limitation in the physics rather than in the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated cross section.

Contents

1	Pion momentum: the range estimator and why it fails	2
1.1	The pion-momentum bias and why p_π is the worst-conditioned observable	2
2	Rejected: extending the pion threshold below 0.175 GeV/c	12
3	Proton-tagged selection, binning and additional smearing	13
3.1	Additional smearing in the proton-tagged observables	15
4	Hadronic mass, model comparison, and estimator surveys	16
4.1	Proton-tagged χ^2 and the A_C correlation	16
4.2	The W binnings are finer than the resolution	17
4.3	What limits $W_{\pi p}$, and what can be done about it	19
4.4	Survey of pion momentum estimators without a magnetic field	21
4.5	Cross-check methods	23
5	Angular observables: the $\cos \theta_\mu$ versus θ_μ forensics	24
5.1	Angular variable: $\cos \theta_\mu$ versus θ_μ	24
5.2	Total flux-averaged cross section and systematic breakdown	28
6	Binning Optimisation, Systematic Breakdown, and Threshold Validation	30
6.1	Momentum-threshold validation	30
6.2	Binning optimisation	30
6.3	Re-check of every released binning on the 2026-09-04 release	31
7	Multi-run overlay, resonance closure, and configuration closure	32
7.1	Multi-run overlay and a resonance-model closure	32
7.2	Consistency with the sibling ν_e CC 1π measurement	34
8	Configuration closure: is COMB the combination of its own inputs?	34
A	Identity closure: unfolding the prediction with itself	37
B	Data/MC normalisation: history of the corrections (superseded record)	39
B.1	Unfolding covariance ($\sim 26\%$ effect)	39
B.2	EXT (cosmic) subtraction for fake data	39
B.3	Opening-angle binning (review item C4)	40
B.4	Generator normalisation (NuWro vs GENIE)	40
B.5	Software trigger (verified to be a no-op)	40
C	NuWro Fake-Data Closure and Cross-Pipeline Validation	40
C.1	Extraction-bias investigation and algorithm equivalence	43

1 Pion momentum: the range estimator and why it fails

1.1 The pion-momentum bias and why p_π is the worst-conditioned observable

The p_π response is the least diagonal of any observable in the analysis. This section quantifies why, because the cause is physical and sets a floor on what the measurement can say about the pion momentum spectrum.

Two separate effects, only one of them corrected. The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the *muon* mass hypothesis and applied to a pion track. Two distinct things follow.

First, the mass hypothesis itself biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is $\sim 17\text{--}19\%$ from the CSDA range scaling, and it is essentially independent of momentum. This effect *is* corrected, though not in the branch: the reco bin definitions in `ccpi_ppi_bin_config_opt.txt` apply a kinetic-energy-preserving $\mu \rightarrow \pi$ mass swap, $E_\mu \rightarrow T \rightarrow E_\pi \rightarrow p_\pi$, before binning. It works; in the lowest true bin $\langle p^{\text{reco}}/p^{\text{true}} \rangle$ goes from 0.888 with the raw branch to 0.976 with the correction.

Second, and not corrected by anything, charged pions interact hadronically before coming to rest. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. This grows rapidly with momentum and comes to dominate.

The reconstructed momentum saturates. Table 1 and Figure 1 make the consequence plain: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at ≈ 0.34 GeV/ c . Above that the estimator carries almost no information about how energetic the pion actually was.

Table 1: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. “raw” is the `candidate_pion_mom_reco` branch; “corr.” additionally applies the $\mu \rightarrow \pi$ mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

true p_π [GeV/ c]	N	$\langle p^{\text{true}} \rangle$	$\langle p^{\text{corr}} \rangle$	raw/true	corr./true
0.175–0.250	528	0.210	0.205	0.888	0.976
0.250–0.300	270	0.276	0.238	0.790	0.863
0.300–0.350	254	0.325	0.253	0.716	0.779
0.350–0.400	265	0.374	0.270	0.664	0.721
0.400–0.500	403	0.443	0.279	0.581	0.629
0.500–0.600	252	0.548	0.307	0.520	0.560
0.600–0.800	272	0.688	0.326	0.441	0.473
0.800–1.200	164	0.951	0.336	0.329	0.353

What that does to the response matrix. Migrating the five analysis bins with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally 6.2% in the last. For the highest true bin ($p_\pi > 0.55$ GeV/ c) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall all the way into the *lowest* one. Every true bin above the first therefore deposits most of its events into reco bin 1, which is exactly the “pile-up into the first bin” seen in Fig. 19 of the analysis note.

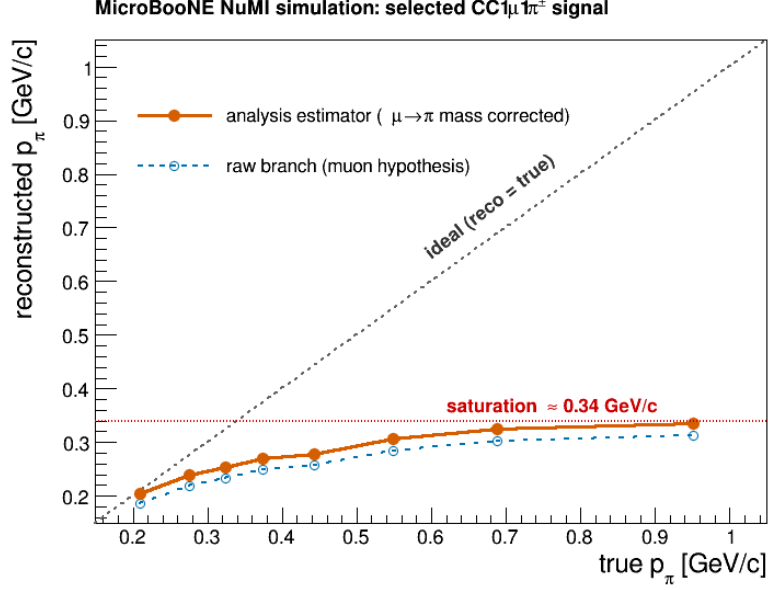


Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled, $\mu \rightarrow \pi$ mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/c: a true 0.95 GeV/c pion reconstructs at the same place as a true 0.55 GeV/c one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

How the BNB $\text{CC}1\pi^\pm$ analysis handles the same problem. The published BNB $\text{CC}1\pi^\pm\text{Xp}$ analysis (internal note v0.91) faces an identical estimator, it too takes “the momentum of pion candidates from range”, and its treatment is worth reproducing here, because it is a solved problem there and, at the time of this study, an open one here.

That analysis does three things we do not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated *golden-pion BDT* and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track “wiggleness” (the standard deviation of the angular difference between directions of adjacent points along the fitted track, so a direct handle on scattering), the number of reconstructed descendant particles, the track-shower score, and the Bragg likelihood ratio. The cost and benefit are quoted explicitly: efficiency falls from 20.3% to 9.5% and purity is essentially unchanged (51.9 $\rightarrow$ 49.5%), but the golden fraction of the selected signal rises from 36.3% to 67.7%. They apply the same logic to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so it uses range only.

Third, and most directly comparable, their binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events *and* the diagonal element of the smearing matrix must exceed 0.68. The resulting pion-momentum edges are 0.10, 0.16, 0.19, 0.22, 0.60 GeV/c with an overflow bin, fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

Set against that criterion our binning does not survive: the diagonal fractions measured above are 94.3, 36.5, 23.1, 10.4 and 6.2%, so *one* of five bins passes 0.68 and the top three fail it by a wide margin. Our edges (0.175–0.550 in five bins, open above) put four boundaries inside precisely the region where the reconstructed momentum has saturated.

Consequences, and what would improve it. This is not a bias on the extracted cross section; the response matrix contains it and the unfolding inverts it, but it is the reason the p_π result leans harder on the regularisation than any other observable (the pre-fix A_C survey of Table 15, superseded in its numbers, showed the momentum observables retaining the least model discrimination through A_C). With 6% diagonal strength in the top bin, the measurement above ~ 0.5 GeV/ c is effectively inferred from the response model rather than measured.

The BNB note therefore supplies a tested recipe, and *all four* of its BDT inputs are already available to us. The Bragg likelihood ratios (`trk_bragg_pion`, `_mip`, `_mu`, `_p`) and the track-shower score are in the ntuples; the multi-pion pion-ID BDT in this analysis already trains on exactly those, and the descendant counts are `pfp_trk_daughters_v` / `pfp_shr_daughters_v`. Wiggleness is there too, under a different name: the PeLEE ntuples carry `trk_avg_deflection_stdev_v` (with companions `_mean_v` and `_separation_mean_v`), which is precisely the standard deviation of the angular deflection between adjacent track points. It is fully populated. It was simply not among the branches `ProcessNTuples` copied forward; it now is.

Wiggleness works, but with the opposite sign to the naive expectation. Testing it directly on truth-matched charged pions in the Run-1 FHC overlay, calling a pion “golden-like” when the mass-corrected range momentum recovers the true momentum to within 20%, and “interacting” when it recovers less than 60%, gives a clear separation, but golden pions are the *wigglier* ones, by about a factor two in the mean (0.0048 against 0.0027). That is not a momentum artefact: it survives binning in track length, with the ratio staying at 0.51–0.63 from 5 cm to 400 cm.

The sign makes physical sense on reflection. A pion that stops travels its full range and scatters violently through its final slow centimetres, so the deflection spread averaged along the track is large. A pion that interacts hadronically is cut short while still fast and never reaches that wiggly end, so its track is straighter. High wiggleness is thus a signature of *having stopped*, exactly the population for which a range-based momentum is meaningful.

As a single cut it is already useful: on this sample the golden fraction is 39.9% before any cut, and requiring `trk_avg_deflection_stdev_v` > 0.001 keeps 86.8% of golden pions while rejecting 48.8% of interacting ones, lifting the golden fraction to 53.0%. A tighter cut at 0.002 gives 57.8% efficiency and 55.7% golden. For reference the BNB’s four-variable BDT reaches 67.7% golden at roughly half the generic-selection efficiency, so one variable recovers a good part of the gain and the remaining three, all of which we have, should close much of the rest.

In increasing order of effort, then: rebin p_π against the BNB’s own criterion (diagonal > 0.68), which on the numbers above means roughly two resolved bins below ~ 0.25 GeV/ c plus a wide bin and an overflow, and can be done today; add a golden-pion cut for the p_π cross section built from the Bragg, track-score and descendant variables already available, accepting the factor-two efficiency loss the BNB analysis accepts, or reproduce their BDT in full, which is now a training exercise rather than a production change. None changes the total or angular results, which do not use this estimator. The first is worth doing before the p_π spectrum is quoted as a shape measurement. *[Subsequently done: the two-bin scheme was adopted (§1.1) and the golden-pion classifier was trained and found insufficient (§1.1).]*

A truncation bug in the generator predictions, found while rebinning

Rebinning p_π required regenerating the generator predictions from the event level, which exposed a separate error in how they were built. The predictions were histogrammed with a closed upper edge, 1.0 GeV/ c for p_π and 3.0 GeV/ c for p_μ , while the corresponding analysis bins are *open*. The overflow was then discarded by `combine_newg4.C`, so the tail above those edges was simply lost.

The consequence is that each generator’s flux-averaged total disagreed with itself depending on which variable it had been binned in. That total is a single number and cannot depend on

the binning variable, which makes this an unambiguous internal check:

generator	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	spread
<i>before</i>						
GENIE	0.790	0.762	0.823	0.823	0.823	8.0%
NuWro	1.198	1.176	1.253	1.253	1.253	6.5%
<i>after</i>						
GENIE	0.823	0.823	0.823	0.823	0.823	0.01%
NuWro	1.253	1.253	1.253	1.253	1.253	0.00%

All four generators were regenerated from their event samples for all three beam configurations, changing only the affected upper edge. The angular observables were already correct; they are complete by construction, since $\cos \theta$ and $\theta_{\mu\pi}$ have no tail to lose, and they now agree with the momenta to better than 0.01%.

The effect on the physics conclusions is modest but not negligible: the generator totals rise by 4–8%, which moves each generator closer to the measured cross section. In the total-cross-section comparison (now the one-bin extraction of Table 17) GENIE went from 1.0σ to 0.9σ below the FHC measurement while NEUT went from 0.8σ to 1.0σ above it, at the uncertainty band then in use; with the current one-bin band these read 0.5σ and 0.7σ . Had the p_π prediction alone been used, the 8% deficit would have appeared as a data–model disagreement in that one observable and nowhere else.

Two practical notes for anyone repeating this. The NEUT and NuWro event readers cannot be run natively: NEUT’s `neutclass` libraries are built against ROOT 5 and NuWro needs its own event library, so both require the SL7 container. And when rewriting histograms into an existing ROOT file, read the inputs fully into memory first, writing while an input file is the active directory silently sends the histogram to the wrong file, which is how a first attempt at this appeared to succeed while leaving the RHC predictions untouched.

Adopted response: a two-bin p_π scheme

The binning has been changed accordingly. The five-bin scheme is replaced by two bins, [0.175, 0.205] GeV/ c and everything above, which raises the migration diagonals from 96/35/21/10/6% (full analysis sample; 94.3/36.5/23.1/10.4/6.2% on Run 1 alone, Table 1) to **91/76%**, clearing the 68% criterion in *both* bins. (These are measured on the analysis sample; the golden-pion study of Sec. 1.1 uses a looser track sample whose diagonals are lower, and the two sets are not interchangeable.)

The boundary is not a free choice dressed up as an optimisation. The worst diagonal is always the upper bin and it falls monotonically as the boundary rises, so the binding constraint is the *width* of the first bin rather than the diagonal itself:

boundary	first-bin width	diagonal, bin 1	diagonal, bin 2
0.185	10 MeV/ c	79.7%	85.4%
0.195	20 MeV/ c	88.8%	80.0%
0.200	25 MeV/ c	90.4%	77.7%
0.205	30 MeV/c	91.0%	75.5%
0.220	45 MeV/ c	93.3%	69.1%
0.250	75 MeV/ c	95.6%	56.1%
0.300	125 MeV/ c	96.3%	39.0%

Both bins clear the criterion anywhere between roughly 0.19 and 0.22, so the choice within that window is governed by bin width rather than by the criterion. 0.205 is the narrowest first bin with precedent: 30 MeV/ c matches the narrowest bin used in the BNB CC1 π analysis. Going narrower trades first-bin diagonal for upper-bin diagonal and at 10 MeV/ c the first bin itself

falls to 80% on only 325 events, which is not a useful cross-section bin; going wider, 0.220 still passes but leaves the upper bin at 69%, with no margin. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and for the proton-tagged selection as well.

A three-bin variant was built and measured before being discarded. The best achievable (same 30 MeV/ c minimum width, edges 0.205 and 0.265) is 91/51/52%, against 91/76% for two bins: the third bin costs 25 points on the worst bin and drops it well below the criterion. Two 30 MeV/ c bins followed by an open bin (0.205, 0.235) is worse still at 91/39/62%, and note that it degrades the *upper* bin too, from 76% to 62%: narrowing the top bin's reco range pushes genuinely high-momentum events out of their own bin. Four bins reach only 91/39/39/42%. Above ~ 0.25 GeV/ c the reconstructed momentum barely varies, so a second boundary there separates almost nothing and merely divides one poorly-measured region into two equally poor halves. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

Closure with the new binning (FHC). The two-bin scheme was run through the full chain, universe construction, Wiener-SVD unfolding, systematics, and closed cleanly on the release of the time (2026-08-17; the current values are those of Table 2 below and of the analysis note):

	σ_{int}	χ^2/ndf	p	total syst.
old five-bin	0.964	1.96/5	0.85	32.3%
new two-bin	0.923	0.14/2	0.93	39.8%

Table 2: Two-bin p_π result for all three beam configurations, $d\sigma/dp_\pi$ in 10^{-38} cm²/(GeV/ c)/Ar. Bin 1 is [0.175, 0.205] GeV/ c , bin 2 everything above. All curves are A_C -smeared, so they live in the same space as the measurement. Generator predictions use the corrected (untruncated) FTE files. Regenerated from the 2026-09-06 release (`report/tools/ppi2bin_tables.py`: closure sidecars and `curves_incl_*_ppi2bin.tsv`); the uncertainty row is the total.

	FHC			RHC			Combined		
	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}
unfolded data	1.96	0.96	0.822	1.79	0.87	0.748	2.06	0.90	0.775
uncertainty	± 1.36	± 0.37		± 1.12	± 0.34		± 1.26	± 0.34	
A_C truth	2.34	0.95		1.71	0.89		2.17	0.90	
uB tune	2.29	0.93		1.72	0.88		2.15	0.89	
GENIE	1.82	0.75		1.29	0.63		1.67	0.67	
GiBUU	3.28	0.94		2.27	0.82		3.03	0.85	
NEUT	3.09	1.18		2.24	1.02		2.87	1.07	
NuWro	2.98	1.12		2.11	0.96		2.74	1.01	
χ^2/ndf	0.11/2 ($p = 0.95$)			0.01/2 ($p = 0.99$)			0.01/2 ($p = 1.00$)		

On the current release all three configurations close, with χ^2 of 0.11, 0.015 and 0.01 on two bins (the RHC value rounds either way in the note's tables) and unfolded-to-truth integral ratios of 0.997, 0.981 and 0.989 (width-weighted unfolded over A_C -smeared truth from the released curves, `curves_incl_*_ppi2bin.tsv`; the χ^2 from Tables 17–19 of the analysis note). The four generators bracket the data in both bins of every configuration, NEUT and GiBUU highest and GENIE lowest, the same ordering as in the other observables; the released A_C for FHC is $\begin{pmatrix} 0.53 & 0.01 \\ -0.51 & 0.87 \end{pmatrix}$ (`A_C_incl_FHC5_ppi2bin.tsv`), whose off-diagonal element is the reminder that the two bins are not independent after merging. The total fractional uncertainty is 50% (FHC), the

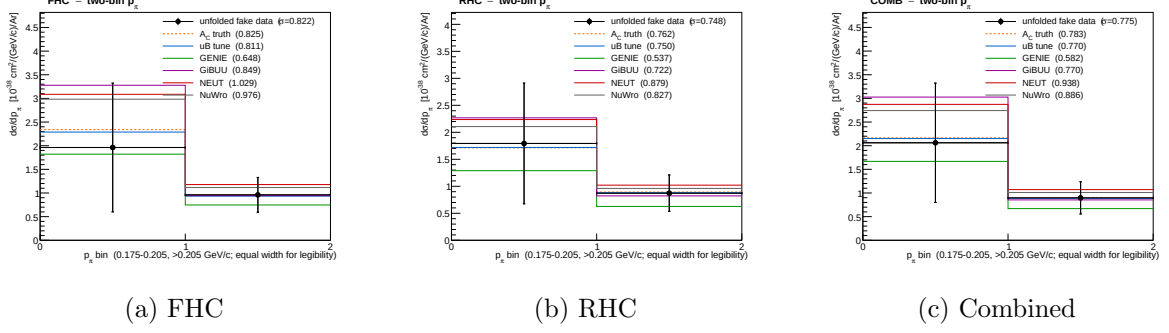


Figure 2: The two-bin p_π cross section, the graphical form of Tab. 2. Unfolded fake data (points, with the full systematic uncertainty) against the A_C -smeared truth, the uB tune and the four generators; every curve is A_C -smeared, so all of them live in the same space as the measurement. **The two bins are drawn with equal width for legibility:** they are 0.030 and 0.795 GeV/c wide, so to scale the first would occupy 4% of the axis. Area on these panels is therefore not proportional to cross section, and the integrated σ is printed on each panel instead. The generators bracket the data in every bin of every configuration; GENIE is generally the lowest prediction and NEUT and GiBUU the highest, and which generator is closest changes from bin to bin.

largest of the inclusive observables, because the two bins each carry a large share of the flux normalisation with little statistical averaging. Figure 3 shows A_C for all three configurations (release 2026-09-06). RHC regularises a little harder than FHC in the first bin (diagonal 0.38 against 0.53, off-diagonal -0.37 against -0.51), and the combined extraction borrows the most (diagonal 0.53, off-diagonal -0.58); the second-bin diagonals are 0.87–0.88 in all three. Despite the different conditioning the three extractions close in normalisation within 2% (the ratios 0.997, 0.981 and 0.989 above), so there is no common normalisation offset associated with how hard A_C smears; an earlier draft quoted a ratio of 1.13 here, which belonged to a superseded fake-data pass and is withdrawn.

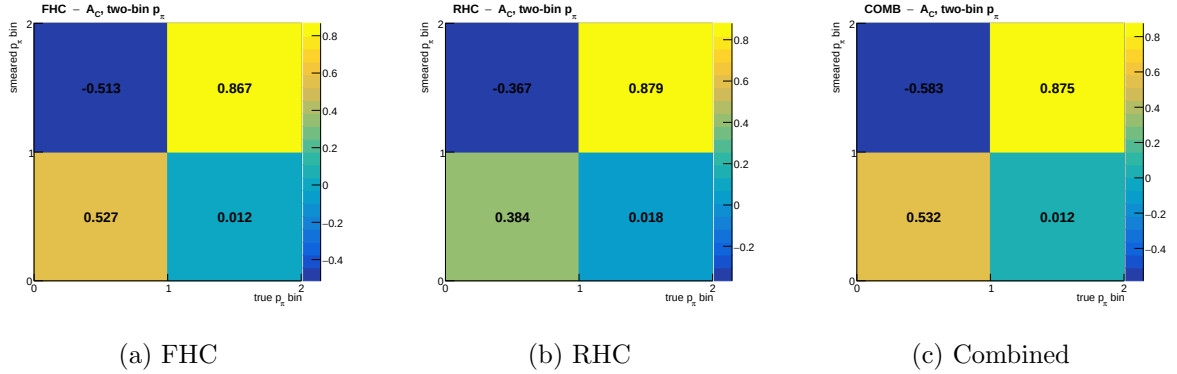


Figure 3: The additional-smearing matrix A_C for the two-bin p_π scheme, in the convention $\hat{x}_i = \sum_j (A_C)_{ij} t_j$ so that a column is a true bin and a row is the smeared bin it feeds. A_C is not normalisation-preserving, so its rows do not sum to unity. The negative entry (true bin 1 feeding smeared bin 2, drawn top-left since the smeared axis runs upward) is the regularisation borrowing between the two bins; it is why the truth curves in Fig. 2 must be A_C -smeared before being compared with the measurement rather than plotted raw.

At 91/76% both bins clear the 68% criterion, so the two-bin scheme makes the inclusive p_π result usable on the standard test rather than merely honest about its limitations. It remains a coarse measurement: two bins cannot resolve a shape, and above ~ 0.34 GeV/c the reconstructed

momentum saturates, so the upper bin is an integral over a wide range rather than a differential point. What the scheme buys is a result whose response is defensible, not a competitive shape measurement.

A golden-pion classifier was trained and tested: it works, and it is not enough

Following the BNB recipe, a golden-pion BDT was trained and evaluated (`macros/golden_pion_train.C`, `macros/golden_pion_compare.C`). The truth label is `mc_end_p < 0.01 GeV/c`, the pion ranged out, joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track-shower score and the descendant count. Training used 126 518 labelled tracks from Runs 1, 2 and 4; the split is by *run*, so all numbers below are on Run 2, which the classifier never saw.

As a classifier it reproduces the BNB benchmark. ROC integral 0.758. At 44.8% efficiency it lifts the golden fraction from 43.1% to 64.9%, against the BNB’s 36.3 $\rightarrow$ 67.7% at roughly 47% efficiency. The variable ranking is dominated by the deflection variables (`wmean`, then `wstd`), with the Bragg ratios contributing essentially nothing, worth noting, since Bragg was expected to carry the stopping signature.

But it does not fix the binning, and cannot. Table 3 gives the migration diagonal in the five analysis bins as the cut is tightened. The first bin is already fine and stays fine; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 3: Golden-pion BDT: efficiency, achieved golden fraction, and the p_π migration diagonal per true bin, on the held-out run. The BNB criterion for a usable bin is a diagonal above 0.68.

BDT cut	eff.	golden frac.	b1	b2	b3	b4	b5
none	100%	43.1%	99.2	23.7	13.2	6.4	5.5
> -0.05	57.0%	58.1%	99.1	29.4	19.7	11.2	7.1
> 0.00	44.8%	64.9%	99.1	33.1	24.5	15.4	9.1
$> +0.10$	24.6%	79.4%	99.1	42.9	36.8	29.3	15.7
$> +0.15$	15.2%	84.5%	99.2	46.9	42.2	36.1	19.5
<i>perfect (truth) golden selection</i>			98.9	36.1	24.1	14.2	8.9

These diagonals are measured on a different sample from the analysis ones, and the two must not be mixed. The table above is built on the labelled *track* sample: every truth-matched pion track passing loose quality requirements (purity > 0.5 , track score > 0.5 , length > 5 cm), from events of any topology, 126 518 tracks in total. The migration diagonals quoted elsewhere in this note, 96/35/21/10/6% for five bins and 91/76% for two, are measured on the *analysis* sample: events passing the full $CC1\mu1\pi^\pm$ selection and satisfying the signal definition, about 10 700 per beam mode. The analysis sample has the better response, because the selection removes exactly the events whose pion is poorly reconstructed. On the track sample the same five-bin scheme reads 99/24/13/6/6% and the two-bin scheme 98/62%.

An earlier draft of this note quoted those track-sample figures as though they were the analysis response, and concluded from the 62% that the inclusive p_π measurement failed the usable-bin criterion. It does not: on the analysis sample the upper bin sits at 76%. The numbers in this table are correct for what they measure, and the conclusions drawn *within* it, which compare rows of the same sample, are unaffected.

The last row is the decisive one, and it must be read against the *efficiency* column rather than against the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives 36.1, 24.1, 14.2, 8.9% in bins 2–5; the classifier at the matching 44.8% efficiency gives 33.1, 24.5, 15.4, 9.1%. They agree bin by bin. The classifier is therefore already performing as well as perfect knowledge of which pions stopped, and no better classifier exists to be built.

Multiple Coulomb scattering was tried as a pion momentum estimator, and as a classifier input. Range saturates because the pion interacts before stopping, which truncates the track. Multiple-Coulomb-scattering (MCS) momentum is inferred from the scattering along whatever track exists and is not truncated by an interaction, so it is the natural candidate to replace range. The CC0 π analysis uses exactly this pairing, range for contained tracks and MCS for exiting ones. Measured on 122 878 truth-matched pion tracks (`trk_mcs_muon_mom_v`, filled for 99.9% of them):

true p_π [GeV/c]	$\langle \text{len} \rangle$	$\langle \text{range/true} \rangle$	RMS	$\langle \text{MCS/true} \rangle$	RMS
0.175–0.250	25 cm	0.834	0.16	4.335	14.1
0.320–0.420	46 cm	0.629	0.23	2.438	7.2
0.550–0.800	71 cm	0.446	0.18	1.423	3.5
0.800–1.500	84 cm	0.319	0.16	1.011	2.1

The trade is stark. In the top bin range is biased to 0.319, which is the saturation, while MCS is essentially unbiased at 1.011; but MCS carries a 210% RMS there and 1400% in the lowest bin. It is unusable as a per-event estimator. The reason is structural and explains why the CC0 π recipe does not transfer: they apply MCS to *exiting* tracks, which are long by construction, whereas our problematic pions are contained but interacting, 25–84 cm long, and MCS needs of order a metre.

The same note uses the *disagreement* between the two estimators as a track-quality BDT input rather than as an estimator. That is well motivated here: a pion that interacts has a truncated range but an untruncated MCS, so the fractional difference $(p_{\text{range}} - p_{\text{MCS}})/p_{\text{range}}$ should flag precisely the events that break the response. Adding it to the golden-pion classifier, it ranks *second of ten* in importance, behind only the mean deflection. It nonetheless changes nothing:

classifier inputs	ROC integral	diagonals at $\sim 44\%$ efficiency
baseline (nine variables)	0.758	99.1/33.1/24.5/15.4/9.1
+ range–MCS difference	0.757	99.0/34.8/24.8/14.8/9.1

The interpretation is that the new variable is highly informative but not *additionally* informative: it is strongly correlated with the deflection variables already present, since both measure whether the track scattered heavily and ended abruptly. High importance rank with unchanged ROC is the signature of a redundant input. This is an independent confirmation of the ceiling argument above: the classifier already extracts what golden-ness has to offer, and the limit is the physics of the label rather than the quality of the classifier.

One tempting variant was deliberately not pursued. Retraining on the label “range is accurate”, $|p_{\text{corr}}/p_{\text{true}} - 1| < 0.2$, would target the quantity we actually care about and would certainly raise the diagonals. It would also be selecting on the answer: keeping events where the estimator happens to be right sculpts the response by construction and inflates the diagonal without measuring anything, the same trap documented for the tight BDT working point above. The golden label is used precisely because it is a physical property of the pion, independent of how well it was reconstructed.

A caveat on the tighter working points. The > 0.15 row looks better than perfect truth selection (46.9, 42.2, 36.1, 19.5%), which is impossible for an honest classifier and is worth understanding rather than quoting. It arises because a hard cut on the classifier preferentially keeps *long* tracks: inside the top true bin, the surviving tracks have a mean length of 120.8 cm against 78.2 cm for all pions, and a mean reconstructed momentum of 0.406 against 0.307 GeV/ c . The cut is thus partly a cut on the reconstructed momentum itself, which raises the diagonal by construction while sculpting the spectrum. The apparent gain is circular and the tight working points should not be used. This is the same failure mode anticipated for track length as a direct input; it enters here through the deflection variables, which correlate with length, even though length itself was excluded from training.

Why, and what is left. Golden pions still reconstruct at $\langle p^{\text{corr}}/p^{\text{true}} \rangle = 0.73$ on this sample. Interaction is therefore not the only problem: the momentum conversion is itself biased. A single flat rescaling does *not* repair it, applying the best global factor drops the first bin from 98.9 to 49.1% and leaves 0/5 bins passing; because the bias is momentum-dependent, exactly as the saturation curve of Figure 1 shows.

Coarser binning helps but does not rescue it either. Collapsing to two bins (0.175, 0.230, ∞) gives 98.7 and 58.3% with perfect golden selection: better, still short of 0.68. Only a single bin; that is, no differential measurement at all, trivially passes.

Conclusion. The comparison is worth stating plainly rather than filing as a partial success. The classifier does what it was built to do and matches the published benchmark, but the p_π differential measurement above ~ 0.23 GeV/ c cannot be made to meet the BNB’s own binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis passes that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all sit *below* our 0.175 GeV/ c threshold, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The realistic options for this analysis are therefore to lower the pion threshold toward 0.10 GeV/ c and resolve the region where the estimator works, to adopt a two-bin-plus-overflow p_π result and say so, or to replace the range-based estimator altogether. The golden-pion BDT is a useful ingredient in any of those, but it is not by itself the fix.

Scope of this conclusion. The failure described in this subsection concerns attempts to recover a finer multi-bin p_π spectrum, and its numbers are for the loose truth-matched track sample. It does not invalidate the adopted two-bin analysis-sample result (§1.1, Table 2), whose migration diagonals are 91% and 76%.

Scope of p_π . Both candidate routes to a finer p_π measurement were tried; as the preceding subsection and the following paragraph show, the golden-pion classifier does not repair the response even at perfect truth-level selection, whereas the proton tag does not change the response appreciably either way. At the two-bin scheme the inclusive selection already satisfies the binning criterion, so p_π need not be withdrawn from it; what it cannot support is the five-bin *shape* measurement originally planned. The other inclusive observables are unaffected: the angular variables and p_μ retain their shape information, and the total cross section does not use this estimator at all.

The proton-tagged route, now measured in all three configurations. Both routes have now been quantified. The golden-pion route is above; the proton-tagged one has been run for FHC, RHC and the combined exposure. The expectation stated in earlier drafts, that requiring a tagged proton “does nothing directly to the pion reconstruction”, turns out to be *correct*: at the same two-bin edges the proton-tagged response is indistinguishable from the inclusive one, marginally worse rather than better. Applying the same migration-diagonal test gives

selection	config.	N_{sig}	bin 1	bin 1 [0.175, 0.205]	bin 2 (> 0.205)	both > 0.68?
inclusive CC1mu1piXp	FHC	1056		91.0%	75.5%	yes
proton-tagged	FHC	469		88.9%	74.2%	yes
CC1mu1pi1p	RHC	621		89.4%	75.1%	yes
	comb.	1090		89.2%	74.7%	yes

measured on the numuMC overlays named in the corresponding `file_properties` lists, selected signal, with the reconstructed momentum mass-corrected exactly as the bin config does it. All four rows clear the criterion in both bins, and the proton-tagged values sit about two points *below* the inclusive ones rather than above them. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

That is the expected result on reflection, and it corrects a claim made in an earlier draft of this note. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion’s own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but it does not change what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response. The measurement now says so directly.

The consequence is that the proton-tagged p_π measurement stands on its own merits, as the pion-momentum cross section of an exclusive final state that also yields W and TKI, and not as a repair of the inclusive one. The inclusive p_π at two bins is equally defensible on the response test.

Table 4: Two-bin p_π on the proton-tagged CC1mu1pi1p selection, all three beam configurations. $d\sigma/dp_\pi$ in $10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$; bin 1 is [0.175, 0.205] GeV/c, bin 2 everything above. All curves are A_C -smeared. The proton-tagged `xsec` configs declare only the tune and the fake data, so no generator rows appear. Regenerated from the 2026-09-06 release (`report/tools/ppi2bin.tables.py`); the uncertainty row is the total.

	FHC		RHC		Combined	
	bin 1	bin 2	bin 1	bin 2	bin 1	bin 2
unfolded data	1.086	0.586	0.386	0.525	0.810	0.551
uncertainty	± 0.836	± 0.250	± 0.796	± 0.222	± 0.851	± 0.221
A_C truth	1.145	0.598	1.056	0.458	1.243	0.518
uB tune	1.139	0.591	1.135	0.469	1.287	0.521
unfolded/truth	0.95	0.98	0.37	1.15	0.65	1.06
σ_{int}	0.498		0.429		0.462	
χ^2/ndf	0.01/2 ($p = 1.00$)		1.21/2 ($p = 0.55$)		0.40/2 ($p = 0.82$)	

All three extractions close, with χ^2 of 0.01, 1.21 and 0.40 on two bins on the current release. The p_π differential cross section satisfies the field’s binning criterion in every bin of every beam configuration here, as it does on the inclusive selection; the two are equivalent on that test, and the proton-tagged version is reported because it belongs to the exclusive final state, not because it is required to make p_π measurable.

The least-populated bin. On the release of 2026-08-17 the RHC first bin unfolded to 1.88 times its A_C -smeared truth; on the current release, after the re-throw, it sits at 0.37 times it (0.386 ± 0.796 against 1.056), with the closure χ^2 of 1.21/2 unremarkable either way. That the same bin swings from $+1.1\sigma$ to -0.8σ between two independent throws is the honest description

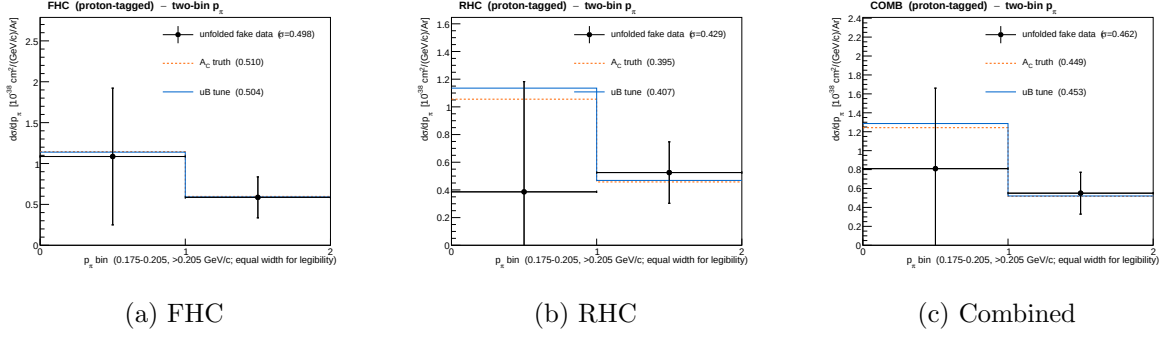


Figure 4: The two-bin p_π cross section on the proton-tagged CC1mu1pi1p selection, the graphical form of Tab. 4. Same edges as the inclusive result of Fig. 2 so the two are directly comparable, and the bins are again drawn equal-width.

of it: the bin holds too few events to say anything, and any real-data value in it should be read with its uncertainty and not its central value.

One caveat remains. The 0.205 GeV/c boundary was independently preferred by the migration-level edge scan run on the proton-tagged selection (`macros/ppi_binning_scan.C`, FHC; `logs/binning_scan/1p_fhc.txt`) as well as by the inclusive scan, so it is not merely inherited. It was not, however, re-optimised through a separate universe, covariance and unfolding production for every candidate boundary, which would need its own universe construction per configuration (~ 1 – 2 h each); that is not expected to change the conclusion, since both bins clear the criterion with margin in all three proton-tagged configurations.

2 Rejected: extending the pion threshold below 0.175 GeV/c

Historical diagnostic record. The extension described here was REJECTED and is not part of the measurement; the analysis threshold is $p_\pi > 0.175$ GeV/c.

The low-momentum bin, added. We have implemented this extension for p_π : a $[0.113, 0.175]$ GeV/c bin is prepended to the pion-momentum binning, lowering the measured phase space to the ν_e threshold. The truth-level signal below 0.175 GeV/c is recovered from the stored signal-component branches (the reconstruction reproduces MC_Signal above 0.175 GeV/c to 0.4%, the residual being the unstored kaon veto); 3323 true signal events populate the new bin. Its selection efficiency is 2.2%; an order of magnitude below the 15–17% of the standard bins (Table 5), as anticipated from the reconstruction turn-on. Despite this, the Wiener-SVD extraction remains well behaved with the bin included: the M_A^{RES} fake-data closure gives $\chi^2/\text{ndf} = 1.41/6$ ($p = 0.97$) and the flux term is unchanged (17% reco, 18.4% unfolded, amplification 1.08 $\times$). The bin therefore adds to the measurement, at the price of a large per-bin uncertainty driven by the low efficiency.

The generator predictions have been regenerated on the six-bin scheme (the ppi histogram uses $p_\pi > 0.113$ while the other observables retain $p_\pi > 0.175$, mirroring the analysis). GENIE, NuWro and NEUT all provide the low bin; GiBUU could not be re-binned as its generator-level event record was not retained, so it is omitted from the extended p_π comparison. Against the (fake) data the generators give $\chi^2/\text{ndf} = 11.1/6$ (GENIE, $p = 0.09$), 7.0/6 (NuWro, $p = 0.33$) and 6.3/6 (NEUT, $p = 0.39$): NEUT and NuWro describe the extended spectrum well, while GENIE is mildly disfavoured, consistent with the known GENIE over-prediction of low-momentum CC $\pi^\pm$ production.

Table 5: Selection efficiency per true p_π bin with the low-momentum bin added. The new $[0.113, 0.175]$ GeV/c bin sits at 2.2%, an order of magnitude below the standard bins.

True p_π [GeV/c]	N_{true}	N_{reco}	Efficiency
0.113–0.175 (new)	3323	74	2.2%
0.175–0.250	3368	526	15.6%
0.250–0.320	2190	367	16.8%
0.320–0.420	2831	479	16.9%
0.420–0.550	2499	410	16.4%
0.550–1.000	2933	429	14.6%

3 Proton-tagged selection, binning and additional smearing

The proton-tagged cut-flow, its per-observable binning study and its A_C factors are collected here rather than in the analysis note. The note reports the extracted proton-tagged cross sections and the reasons that sample is secondary scope; the material below is the supporting detail, which was taking more space in the note than a secondary result warrants.

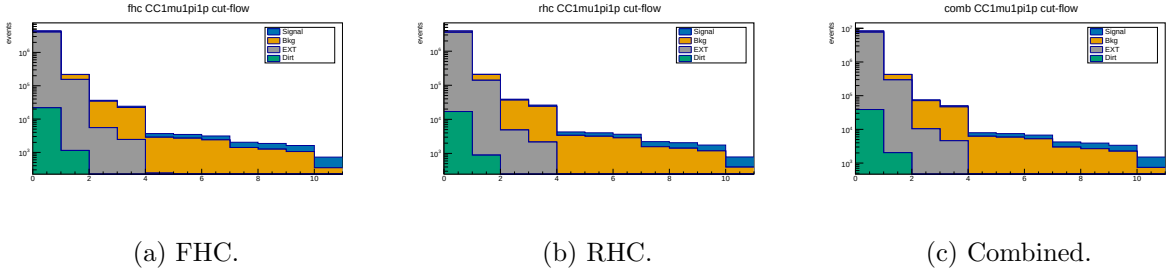


Figure 5: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure (log scale; MC-prediction stack, Okabe-Ito: signal blue, background orange, EXT grey, dirt green). The last stage, IdentProton, is the leading-proton identification that defines the subsample.

Table 6: Proton-tagged (**CC1mu1pi1p**) cut-flow yields at full exposure for FHC, RHC and combined (MC prediction; data blind). Stages None–Nonprotons are the inherited inclusive selection with the proton-tagged signal definition; the final IdentProton stage adds the leading-proton identification. Signal/Bkg from the ν_μ overlays POT-scaled per run, reprocessed with the proton-tagged selection and the full central-value weight under the safe-weight rule (**reprocess_cutflow_cf_1p.sh**, 2026-09-05); EXT from the per-run beam-off samples scaled by the run gate ratios; dirt from the GENIE NuMI dirt overlay (**macros/cutflow_yields_1p.C**). The final row agrees with the proton-tagged cut-and-count check of the analysis note (Table 21 of the analysis note). Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
<i>FHC (Runs 1,2,4,5; 8.857×10^{20} POT)</i>							
None	3 295	270 438	4 049 292	21 864	4 344 889	100.0	0.08
InFiducialVol	2 763	61 976	153 533	1 146	219 418	83.9	1.3
Topological	2 183	28 864	5 409	125	36 580	66.2	6.0
MuonCandidate	1 977	19 865	2 363	84	24 288	60.0	8.1
ContainedPion	830	2 608	223	16	3 676	25.2	22.6
MuonIn3Planes	790	2 472	185	11	3 458	24.0	22.9
PionIn3Planes	733	2 230	144	4.4	3 112	22.3	23.6
ShowerCut	613	1 294	108	3.8	2 019	18.6	30.4
OpeningAngle	609	1 207	41	1.2	1 858	18.5	32.8
Nonprotons	550	1 031	30	0.8	1 612	16.7	34.1
IdentProton	376	338	8.5	0.3	723	11.4	52.1
<i>RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} POT)</i>							
None	3 427	288 891	3 560 975	16 958	3 870 250	100.0	0.09
InFiducialVol	2 886	67 416	139 122	889	210 312	84.2	1.4
Topological	2 318	31 870	4 767	97	39 052	67.7	5.9
MuonCandidate	2 111	21 962	2 090	65	26 227	61.6	8.0
ContainedPion	882	3 181	191	12	4 267	25.7	20.7
MuonIn3Planes	842	3 015	158	8.3	4 023	24.6	20.9
PionIn3Planes	784	2 739	121	3.4	3 648	22.9	21.5
ShowerCut	647	1 480	86	2.9	2 216	18.9	29.2
OpeningAngle	643	1 384	33	0.9	2 060	18.8	31.2
Nonprotons	573	1 162	23	0.6	1 759	16.7	32.6
IdentProton	391	388	7.1	0.2	786	11.4	49.8
<i>Combined FHC+RHC (1.994×10^{21} POT)</i>							
None	6 721	559 329	7 610 266	38 822	8 215 139	100.0	0.08
InFiducialVol	5 649	129 392	292 655	2 035	429 731	84.0	1.3
Topological	4 501	60 734	10 176	221	75 632	67.0	6.0
MuonCandidate	4 088	41 826	4 453	148	50 515	60.8	8.1
ContainedPion	1 712	5 790	414	28	7 943	25.5	21.6
MuonIn3Planes	1 632	5 487	343	19	7 481	24.3	21.8
PionIn3Planes	1 518	4 969	266	7.8	6 760	22.6	22.5
ShowerCut	1 260	2 774	194	6.7	4 235	18.7	29.7
OpeningAngle	1 251	2 591	74	2.2	3 918	18.6	31.9
Nonprotons	1 123	2 193	53	1.5	3 371	16.7	33.3
IdentProton	768	726	16	0.5	1 509	11.4	50.9

Table 7: Released binning of the six W/TKI proton-tagged observables, read from the data release (`curves_1p_*.tsv`). The two mass variables keep six equal-population bins chosen from the validated true-signal distributions, with the last bin open at truth level (the upper edge listed is the plotting boundary); the four TKI variables use the coarsened two-bin scheme adopted after the six-bin scheme was found to sit inside the resolution (§4.2). $W_{\pi p}$ brackets the $\Delta(1232)$ peak but its differential shape is withdrawn. The five inherited kinematic observables (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) use the inclusive-analysis binning of Appendix A of the analysis note unchanged.

Observable	Definition	Bins	Released bin edges
$W_{\pi p}$	$M(\pi p)$, direct	6 (withdrawn)	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9 GeV/ c^2
W_{had}	calorimetric	6	0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74 GeV/ c^2
δp_T	$ \vec{p}_T^\mu + \vec{p}_T^{\text{had}} $	2	0, 0.3, 2.5 GeV/ c
$\delta\alpha_T$	TKI angle	2	0, 120, 180°
$\delta\phi_T$	TKI angle	2	0, 50, 180°
p_n	inferred nucleon momentum	2	0, 0.375, 2 GeV/ c

3.1 Additional smearing in the proton-tagged observables

Because the Wiener–SVD additional-smearing matrix A_C is not norm-preserving, each observable’s predictions are rescaled by a different factor when they are brought into the measurement space. Table 8 gives that factor, measured as the ratio of the A_C -smeared to the truth-level generator integral, for the six proton-tagged observables. Its dominant behaviour is set by the response and the binning; the exact factor is prediction-dependent, and for the tested generators GENIE and GiBUU agree to a few per cent throughout.

On the current release the six proton-tagged observables are suppressed by similar amounts, 0.61–0.79 (Table 8): the two-bin TKI variables and $W_{\pi p}$ sit at 0.70–0.79 and the six-bin W_{had} , the most affected, at 0.61. Sharpness alone does not drive the suppression: $W_{\pi p}$ is the sharpest distribution measured here (the ratio of its largest to smallest bin is ~ 36) yet is suppressed no more than the TKI variables, because A_C removes structure that the *response* cannot resolve and $W_{\pi p}$ is built from directly measured pion and proton momenta with equal-population bins that track the $\Delta(1232)$ peak. No observable is enhanced by A_C any more: the pre-fix set, in which δp_T was suppressed to 0.43 and p_n amplified to 1.7, was an artefact of the uniform-grid regularisation (footnote to the table), and the caveat that once attached to those two panels no longer applies. The suppression is applied identically to data, tune and generators, so model comparisons read off any panel are made on equal footing.

Table 8: Mean A_C row sum for the proton-tagged observables (FHC), from the released matrices `A_C_1p_FHC5_*.tsv`, release 2026-09-06; equal to the ratio of the A_C -smeared to the truth-level integral for a flat prediction. The inclusive observables are given for scale.

Observable	mean A_C row sum
p_n	0.79
$\delta\alpha_T$	0.79
$W_{\pi p}$	0.75
$\delta\phi_T$	0.72
δp_T	0.70
W_{had}	0.61
<i>Computed from the released matrices (<code>A_C_1p_FHC5_*.tsv</code>) after the width-aware regularisation and the removal of the duplicated A_C. The values span 0.61–0.79, a range of 0.18. The pre-fix set (0.88, 0.78, 0.63, 0.61, 0.43, 1.71) spanned a factor of four and amplified p_n above unity; the ordering of the generator χ^2 can no longer be attributed to differences in smearing.</i>	

4 Hadronic mass, model comparison, and estimator surveys

4.1 Proton-tagged χ^2 and the A_C correlation

[SUPERSEDED — DO NOT QUOTE.] Withdrawn; since re-extracted.

Every χ^2 in this subsection compares a measurement of the transverse variables taken about the *detector z axis* with generator predictions taken about the *neutrino direction* (Sec. 5.2 of the analysis note). They are not the same variables, and the numbers below therefore do not measure model agreement. The proton-tagged transverse observables have since been re-extracted in the beam frame, with the width-aware regularisation, a single application of A_C and full detector covariance; those are the current results, and the values in this subsection are retained only as the record of what the coordinate error looked like. The size of the mismatch is not subtle. Under the two conventions the true δp_T distribution over the current bin edges runs 20.7/17.1/14.7/15.2/16.1/16.3% about z against 80.8/10.8/4.5/2.5/1.2/0.2% about the neutrino; the generator predictions sit at 77.3/12.3/5.1/3.2/1.9/0.3% (GENIE) and 80.9/10.1/4.4/2.7/1.6/0.3% (NEUT), i.e. they agree closely with the beam-frame truth and not at all with what was measured against them. The large χ^2 values below are that definitional mismatch, not physics. That the generators reproduce the beam-frame truth this well is also a useful independent check of the correction itself.

The bin edges are affected as well as the χ^2 : they were chosen as equal-population quantiles of the uncorrected variable, which is why the old occupancies above are flat. They are being re-derived along with the re-extraction, and the conclusion drawn here — that the χ^2 ordering tracks $|A_C - 1|$ — must be re-established from scratch afterwards rather than assumed to survive.

Table 9 repeats the comparison for the six proton-tagged observables (FHC), with the measured additional-smearing factor of Sec. 3.1 in the last column. Ordering the rows by $|A_C - 1|$ makes an unambiguous pattern visible: the generator χ^2 grows monotonically with the distance of A_C from unity, and does so in the same direction for every model.

Observable	A_C	$ A_C - 1 $	typical generator $\chi^2/6$
W_{had}	0.69	0.31	4.1
$W_{\pi p}$	0.77	0.23	1.6
$\delta\alpha_T$	0.71	0.29	7.1
$\delta\phi_T$	0.80	0.20	15.5
δp_T	0.67	0.33	70
p_n	0.69	0.31	76

[*SUPERSEDED* — this paragraph and the next were written before the beam-axis correction and the removal of the duplicated A_C ; the χ^2 they discuss compare observables computed about the wrong axis, and the θ_μ factor-9 reduction was the double- A_C bug (change log). Retained as record only; current comparisons are the proton-tagged tables of the analysis note.] This argument was made with the pre-fix A_C and does not survive. With the uniform-grid regularisation the smearing factors ranged from 0.43 to 1.71, and p_n was *amplified* by 71%, which is not something a physical smearing matrix should do; on that set $|A_C - 1|$ did order the generator χ^2 , and the ordering was read as a smearing artefact. With the width-aware regularisation the factors cluster at 0.67–0.80 and $|A_C - 1|$ is flat to within 0.13 across observables, while the generator χ^2 still spans 1.6 to 76 per six bins. A flat quantity cannot order a varying one, so the artefact explanation was withdrawn on that pass.

That is the expected behaviour rather than a surprise. δp_T and p_n probe the nuclear initial state and final-state interactions directly, and generators differ most there; the previous A_C was flattening precisely the observables built to discriminate. The same reassessment applies to $\cos\theta_\mu$ (Sec. 11.8 of the analysis note), where reparameterising to θ_μ reduced the NEUT and NuWro χ^2 by a factor ~ 9 on identical data. The closure column is excellent throughout ($\leq 0.46/6$), so the chain itself is sound; it is the *model comparison* in δp_T and p_n that should not be read as discrimination until the binning and response of those two observables are revisited.

Table 9: [**SUPERSEDED — DO NOT QUOTE.**] These values were computed about detector z while the predictions used the neutrino direction, so measured and predicted were different observables; they additionally carry the double application of A_C . Retained only as the record of what the beam-axis error looked like. χ^2/ndf of the unfolded fake data against each A_C -smeared prediction for the proton-tagged observables (FHC), with the additional-smearing factor A_C .

Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro	A_C
W_{had}	0.86/6	2.50/6	4.51/6	3.87/6	3.46/6	4.71/6	0.69
$W_{\pi p}$	0.19/6	1.65/6	2.59/6	1.11/6	0.85/6	1.87/6	0.77
$\delta\alpha_T$	0.67/6	3.46/6	7.41/6	6.74/6	7.27/6	6.83/6	0.71
$\delta\phi_T$	4.02/6	5.96/6	12.22/6	15.40/6	17.60/6	16.82/6	0.80
δp_T	1.87/6	6.37/6	44.43/6	69.31/6	91.61/6	76.04/6	0.67
p_n	0.19/6	4.93/6	39.79/6	74.32/6	108.00/6	83.83/6	0.69

4.2 The W binnings are finer than the resolution

$W_{\pi p}$ shows the least model tension of the proton-tagged observables, and it was also the worst-conditioned A_C in the analysis before the regularisation fix ($s_2/s_1 = 0.022$, against 0.2–0.25 for the rest). Those two facts have a common cause, and it is not physics.

The bin edges were chosen for equal population, six bins of roughly equal content, without reference to the resolution. Measured on selected signal, FHC Runs 1/2/4:

	$W_{\pi p}$	W_{had}
residual RMS [GeV/ c^2]	0.142	0.283
robust half-width (16–84%)	0.101	0.177
bias (mean reco – true)	–0.111	–0.032
narrowest analysis bin	0.040	0.100
widest analysis bin	1.430	1.300
width ratio	35.7	13.0

The second and third $W_{\pi p}$ bins are 40 MeV/ c^2 wide against a resolution of 101–142 MeV/ c^2 : they are two to three times narrower than the detector can resolve. The bias alone, –111 MeV/ c^2 , is larger than three of the six bin widths. The migration diagonals follow directly, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin at 2.3% is essentially unmeasured.

This is the same failure as the five-bin p_π scheme, and for the same underlying reason. $W_{\pi p}$ is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §1.1; its resolution cannot be better than that of its inputs.

An exhaustive edge scan, the same procedure and criterion used for p_π , gives:

scheme	diagonals	verdict
current, six bins	87.6/22.3/11.7/12.0/10.0/2.3%	1 of 6 usable
best two bins (split 1.15)	83.2/77.9%	both usable
best three bins	83.2/49.7/45.9%	fails
best four bins	83.2/45.0/28.2/28.2%	fails

As with p_π , two bins is not merely the best option but the only one that satisfies the criterion, and the first bin is thin (137 selected signal events, against a floor of 100). The implication for the model comparison is direct: the low χ^2 quoted for $W_{\pi p}$ on that pass reflects a binning that cannot resolve the differences between generators, not agreement between them. The same test should be applied to W_{had} , whose upper bins are also narrower than its 177 MeV/ c^2 half-width, before either W result is quoted as a shape measurement. *[Open: the W_{had} edge scan has not been run; W_{had} is released at six bins with its closure and generator overlays, and its shape should be read with this caveat (analysis note, Limitations).]*

Adopted scheme: two bins split at 1.15 GeV/ c^2 . The boundary is more tightly constrained than it was for p_π , where anything between roughly 0.19 and 0.22 worked. Here only one choice satisfies both the diagonal criterion and the 100-event floor:

boundary [GeV/ c^2]	bin 1	bin 2	$N(\text{bin 1})$	
1.13	76.5%	88.0%	51	below the event floor
1.15	83.2%	77.9%	137	adopted
1.17	86.8%	66.8%	266	bin 2 just fails
1.19	87.6%	56.0%	468	fails
1.23	91.7%	37.6%	1114	fails

Moving the boundary up buys population in the first bin at the direct expense of the second: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about ± 0.01 GeV/ c^2 wide. That narrowness is itself a consequence of the resolution being comparable to the full width of the peak region.

Two consequences should be carried with the result. First, **the first bin is thin**, 137 selected signal events against a floor of 100, so its statistical uncertainty will be large once systematics are folded in; the p_π first bin, with 469 events, carried 42–53% on that pass and 77% on the current release (Table 4). Second, if that proves too sparse in practice, 1.17 is the natural fallback: it

doubles the population for a bin-2 diagonal of 66.8%, marginally below threshold, which is a defensible trade to state explicitly rather than a criterion to fail silently.

The scheme is implemented in `configs/ccpi1p.Wpipr_bin_config.2bin.txt`, with the reasoning recorded alongside it in a companion `.README`. That test has since been made, and it is what settled the question: the two-bin scheme meets the migration criterion but leaves only about 137 selected signal events in the first bin, which the Wiener filter then suppresses, driving A_C towards rank one for statistical rather than algebraic reasons. Widening the first bin restores its population at the cost of the upper-bin diagonal, and no scanned edge satisfies both requirements at once.

The apparent $W_{\pi p}$ agreement therefore does *not* survive a binning the detector can resolve, and the differential shape is withdrawn: at this exposure the proton-tagged sample supports an integrated cross section. W_{had} carries the same limitation, since it depends on the same pion momentum.

4.3 What limits $W_{\pi p}$, and what can be done about it

The two-bin $W_{\pi p}$ scheme of §4.2 was built and run. It does not work, for a reason distinct from the six-bin failure, and the investigation that followed settles what the pion momentum can and cannot deliver.

The two-bin extraction fails on statistics. The *response* behaves exactly as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at $\chi^2 = 1.17/2$ ($p = 0.56$). The *extraction* nonetheless carries almost no shape information, because A_C collapses:

(*Matrices of the 2026-08 pass in which the $W_{\pi p}$ two-bin scheme was run, written with true bins as rows; superseded. The released inclusive p_π matrix is that of Fig. 3.*)

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}$, first bin $N = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	0.031
p_π proton-tagged, $N = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
p_π inclusive, $N = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

The top-left entry is 0.018: A_C passes under two per cent of the first bin and takes -0.41 from the second. This is a rank-one collapse of the kind described in §5.1, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes, $137 \rightarrow 0.031$, $469 \rightarrow 0.577$, $1056 \rightarrow 0.528$, follows population rather than anything geometric.

$W_{\pi p}$ is therefore caught between two requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfold (1.19 gives 468 events, matching the p_π case that works) fail the criterion at 56%.

No binning escapes it. The conclusion does not rest on the one scheme that was tried. Scanning *every* two- and three-bin scheme with edges between 1.12 and 1.70 GeV/ c^2 (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

two-bin edge	N_1/N_2	diag ₁ / diag ₂	
1.15	137/3367	0.832/0.779	first bin too sparse
1.18	351/3153	0.866/0.611	
1.24	1298/2206	0.930/0.347	upper bin unusable
1.27 (median)	1744/1760	0.951/ 0.261	
1.45	2966/538	0.993/0.030	

The trend exposes the mechanism. Reconstructed $W_{\pi p}$ is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§1.1), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag_1 climbs from 0.786 to 0.999 across the scan while diag_2 collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better — the third-bin diagonal never exceeds 0.347.

At the current exposure $W_{\pi p}$ does not support a shape measurement and is quoted as an integrated cross section unless the pion momentum improves. That integrated number is a one-bin extraction rather than a regularised unfolding: with a single bin there is nothing to unfold, since the response is 1×1 , A_C is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 12.3 of the analysis note, and integrated over the full range it is the same quantity as the total proton-tagged cross section. The corresponding integrated generator predictions are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

Six-bin $W_{\pi p}$ closure, diagnostic only. For completeness the six-bin extraction was carried through the full chain in all three configurations (detector variations included, 2026-09-06 release). It closes against the A_C -smeared truth, but that closure cannot restore shape information the response does not contain (the top bin is 2% diagonal), so these numbers are a check of the machinery and *not* a result; they were removed from the analysis note’s result tables for that reason. The six-bin generator predictions, rebinned from the earlier three-bin ones with integrals conserved (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT $0.783 \times 10^{-38} \text{ cm}^2/\text{Ar}$, FHC), exist in the release for the same diagnostic purpose.

Table 10: Six-bin $W_{\pi p}$ fake-data closure, diagnostic only (withdrawn as a differential result). σ_{int} in $10^{-38} \text{ cm}^2/\text{Ar}$; columns as in the note’s proton-tagged closure tables.

Config	σ_{int}	unf./truth	χ^2/ndf
FHC	0.510	1.06	0.52/6
RHC	0.305	0.88	2.56/6
Combined	0.390	0.92	1.60/6

The limitation is the pion momentum, and only the pion momentum. Correlating the $W_{\pi p}$ residual against each input residual isolates the cause:

input	r	W residual RMS when...	
pion momentum	+0.726	all events	0.142
proton momentum	+0.012	pion momentum well reconstructed	0.061
pion $\cos \theta$	−0.029	proton momentum well reconstructed	0.130
proton $\cos \theta$	−0.032	both well reconstructed	0.038

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton buys almost nothing, and the angles are irrelevant. A single defect propagates through the whole analysis: the hadronic interaction that truncates the pion track forces p_π to two bins, and forces $W_{\pi p}$ out of a shape measurement altogether.

Reparameterising to W^2 does not help. A monotonic transformation with mapped edges leaves the migration matrix unchanged, which was verified directly: splitting at $1.15 \text{ GeV}/c^2$ in W and at $1.15^2 = 1.3225 \text{ GeV}^2/c^4$ in W^2 gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together, $\sigma(W^2) = 2W\sigma(W)$, so σ/width is invariant at about 2 either way. This is the same lesson as θ_μ versus $\cos \theta_\mu$ (§5.1).

4.4 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism, that each measures what the pion did *before* it interacted:

estimator	$\langle \text{reco/true} \rangle$, top bin	RMS	note
range (analysis)	0.319	0.16	saturates near 0.34 GeV/ c
calorimetric energy	0.383	0.31	also truncated, and $2\times$ noisier
MCS	1.011	2.1	unbiased, unusably noisy on short tracks
golden-pion selection	0.327	0.18	no gain above 0.8 GeV/ c

Calorimetry does not escape the truncation. Deposited charge stops at the interaction point just as range does, so $\langle \text{calo/true} \rangle$ falls from 1.136 to 0.383 across the momentum range while the RMS is three to five times that of range (1.007 to 0.313, against 0.157 to 0.158). It is the same measurement with more noise.

The golden-pion selection is weaker than previously described. Splitting the sample by the truth label shows that golden pions are not well reconstructed either, and that their advantage disappears exactly where it is needed:

true p_π [GeV/ c]	golden fraction	golden $\langle \text{range/true} \rangle$	non-golden
0.175–0.250	77.8%	0.873	0.699
0.320–0.420	44.1%	0.705	0.570
0.550–0.800	21.4%	0.507	0.429
0.800–1.500	17.2%	0.327	0.317

Above 0.8 GeV/ c the golden and non-golden values are identical to within one per cent. The reason is physical: a 1 GeV/ c pion has a CSDA range of several metres in argon and cannot stop inside a 2.5 m TPC unless it has already lost most of its energy through processes that truncate or break the track. “Golden” at high momentum therefore selects a pathological subsample rather than a clean one, which is why the classifier of §1.1 reaches a ceiling: above ~ 0.5 GeV/ c there is no clean subsample to find.

A multivariate regression fixes the bias but redistributes the resolution. A BDTG regression on p_π^{true} , using range, MCS, calorimetry, length, the three deflection variables, the four Bragg likelihoods, track score and daughter count, was trained on Runs 1 and 4 and evaluated on the held-out Run 2. It largely removes the saturation in the mean, $\langle \text{pred/true} \rangle$ running 1.196 to 0.722 against 0.835 to 0.319 for range, at comparable RMS. Rebuilding $W_{\pi p}$ with it:

pion estimator	W bias	W RMS
range (analysis)	−0.127	0.144
BDTG regression	− 0.005	0.119
true p_π (ceiling)	+0.003	0.046

The bias falls by a factor of 25, which matters independently of binning: a -127 MeV/ c^2 shift distorts the whole W distribution. The resolution improves only from 0.144 to 0.119 against a ceiling of 0.046, so the regression recovers about a quarter of what is available.

Its effect on the binning is a redistribution rather than a rescue. The six-bin diagonals move from 98/30/19/19/13/6% to 63/38/33/43/52/56%: the top bin improves nearly tenfold, but the first bin degrades from 98% to 63%, and in the two-bin scheme the worst bin falls from 73% to 58%. That degradation is regression to the mean, the estimator being pulled toward the sample centre, and it damages the low- W region where the resonance physics sits. The same pattern

appears in p_π itself, where the first bin falls from 99.2% to 60.6% while the top bin rises from 0.5% to 33.2%.

Two objections stand against adopting it. The estimator would be MC-trained and sits upstream of everything, so its model dependence becomes a systematic on every result rather than on one cut. And the trade it offers moves measurability out of the region of interest. It is not adopted here.

What would actually work. The ceiling measurements above bound what is available: a perfect pion momentum would give $W_{\pi p}$ a resolution of 0.046, a factor of three better than the regression achieves and enough to support several bins. The information is carried away by the interaction products, so the approach that attacks the cause is to recover their energy: the pion and its daughters treated as a mini-calorimeter, using the per-PFP calorimetric energies and daughter links already present in the input files. For the proton-tagged sample there is a second, independent handle in transverse kinematic imbalance, where the muon and proton constrain the pion transverse momentum without using the pion’s own reconstruction at all, at the cost of Fermi-motion smearing of order 250 MeV/c. The transverse-imbalance route has not been tried; the daughter-recovery route was tested next.

Recovering the interaction products does not work either. The ceiling measurements above locate the missing information in the secondaries produced when the pion interacts, so the natural remedy is to treat the pion and its products as a single calorimeter. This was tested. PeLEE carries no parent index, so association is geometric, by PFP start point relative to the pion track end, optionally restricted to the generation immediately below the pion in the Pandora hierarchy. The hierarchy restriction matters: at a 40 cm radius the unrestricted sum collects 0.371 GeV in the lowest momentum bin, where only 0.034 GeV is missing, so it is dominated by unrelated activity.

With the hierarchy cut applied, on Run 1:

true p_π [GeV/c]	$\langle \text{KE missing} \rangle$	$\langle \sum E \text{ of daughters within } R \rangle$ [GeV]			
		5 cm	10 cm	20 cm	40 cm
0.175–0.420	0.034	0.023	0.026	0.029	0.037
0.420–0.800	0.246	0.040	0.048	0.055	0.062
0.800–1.500	0.659	0.056	0.074	0.091	0.104

“KE missing” is the pion’s true kinetic energy less the calorimetric energy of its own track. In the top momentum bin 659 MeV is unaccounted for and the visible daughters supply 104 MeV, of which about 37 MeV is the baseline seen even in the lowest bin where almost nothing is missing. Genuine recovery is therefore of order 10% at high momentum and 25% in the middle bin.

It also does not change the binning assessment, which was tested directly rather than inferred from the bias. Adding the daughter energy improves $\langle \text{reco/true} \rangle$ everywhere, the top bin moving from 0.373 to 0.470 and the first bin becoming nearly unbiased at 0.953, and it roughly triples the upper migration diagonals. Neither is enough: the best upper diagonal rises from 5.5% to 15.7%, against the 68% required. In the adopted two-bin scheme it is a wash and arguably a loss, the worst bin improving from 61.4% to 71.0% while the first degrades from 97.7% to 82.4%, the same regression-to-the-mean behaviour seen with the multivariate estimator. Since both bins already clear the criterion with plain range, nothing is rescued and the daughter association would be imported as a new systematic for no gain. **The two-bin choice for p_π is therefore robust against every estimator tested**, which is a stronger statement than the binning study alone could make.

That recovery is far too little to change the measurement, and the reason is physical rather than algorithmic. Pion absorption on argon, $\pi A \rightarrow NN \dots$, deposits much of the energy in nucleons, roughly half of which are neutrons and therefore invisible in liquid argon; the charged

products are frequently short and fall below the threshold to be reconstructed as separate particles. The energy is not merely unassociated, it is largely undetected. Applying the daughter sum would move $\langle \text{reco/true} \rangle$ in the top bin from 0.32 to about 0.42, still less than half of truth.

Where this leaves the pion momentum. Five approaches have now been measured: range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that materially improves anything, removing the $W_{\pi p}$ bias, and it does so at the cost of model dependence and of the low- W region.

One avenue remains untried and is independent of the pion’s own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/ c , which is comparable to the present errors rather than better than them, so it is worth measuring but should not be expected to transform the measurement. Failing that, the honest conclusion is that p_π is a two-bin measurement and $W_{\pi p}$ an integrated one at this exposure, and that both are limited by physics rather than by analysis choices.

4.5 Cross-check methods

Provenance. The D’Agostini cross-check of this section was re-evaluated on the 2026-09-06 release (per-run beam-off, re-thrown fake data, five-bin $\cos \theta_\pi$); Table 12 and the numbers quoted around it are from that run. The closure figures of the other methods were evaluated on the 2026-08-31 release and not re-run; the comparison they support is a property of the operators and is unaffected by the re-throw.

Seven additional methods are run in parallel:

Table 11: Unfolding methods.

Label	Description
<code>wiener_svd</code>	Wiener-SVD (primary result)
<code>bin_by_bin</code>	Efficiency correction only; no migration unfolding
<code>ibu</code>	D’Agostini iterative Bayesian, 4 iterations
<code>tikhonov</code>	Tikhonov-regularised least-squares, $\tau = 1.0$
<code>roo_bayes</code>	RooUnfoldBayes, 4 iterations
<code>roo_svd</code>	RooUnfoldSvd
<code>roo_invert</code>	RooUnfoldInvert (direct matrix inversion)
<code>roo_binbybin</code>	RooUnfoldBinByBin

The D’Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C , its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 12 compares the two integrated totals for all three configurations: the D’Agostini result sits above the Wiener-SVD total by 31% (FHC), 29% (RHC) and 23% (Combined), equivalently A_C suppresses the quoted Wiener-SVD integral by 23%, 23% and 19%, and it varies by $\leq 0.3\%$ across 1, 2 and 4 iterations. The D’Agostini integral is flat across the five observables to 1.2% (FHC, 1.017–1.028), 0.9% (RHC, 0.930–0.939) and 0.5% (Combined, 0.972–0.976), against Wiener-SVD spreads of 24%, 35% and 34%, directly confirming that the observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 20 of the analysis note) is the A_C smearing rather than a physical or acceptance effect.

Table 12: Wiener-SVD vs. D’Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five inclusive observables (p_μ , two-bin p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$; 10^{-38} cm²/Ar), on the 2026-09-06 release. D’Agostini carries no A_C smearing, so it recovers a 23–31% higher total, and it is flat across observables (spread 1.2% FHC, 0.9% RHC, 0.5% combined) where the Wiener-SVD integrals spread by 24–35%, confirming that the observable-to-observable variation of σ_{int} in Table 20 of the analysis note is the A_C smearing rather than a physical or normalisation effect. The last column is the change of the D’Agostini integral between 1 and 4 iterations.

Config	$\langle\sigma\rangle_{\text{WSVD}}$	$\langle\sigma\rangle_{\text{DAg}}$	DAg/WSVD	DAg 1→4 iter
FHC	0.782	1.022	1.31	$\leq 0.3\%$
RHC	0.723	0.933	1.29	$\leq 0.3\%$
Combined	0.790	0.974	1.23	$\leq 0.1\%$

5 Angular observables: the $\cos\theta_\mu$ versus θ_μ forensics

5.1 Angular variable: $\cos\theta_\mu$ versus θ_μ

The muon-angle distribution is measured in both $\cos\theta_\mu$ and θ_μ . The two span the same full range but are different partitions of it: the θ_μ edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos images of the $\cos\theta_\mu$ edges (0.45, 0.64, 0.86, 1.10 rad; Table 15 of the analysis note), and they behave very differently under the Wiener-SVD additional smearing A_C , and the choice matters for how the measurement compares to models.

In $\cos\theta_\mu$ the signal is sharply forward-peaked: the last two bins ($\cos\theta_\mu > 0.8$) hold most of the rate, and A_C suppresses the integral of any truth-level prediction strongly (row sums 0.17–0.39 in FHC on the current release, against 0.66–0.91 for p_μ). Because the same A_C is applied to the data, the tune and every generator, they are compared on equal footing; the 2–3× deficit of the generators reported in earlier drafts was the duplicated A_C (change log), not a property of the comparison.

An earlier version of this note attributed both effects to the sharpness of the peak, A_C smearing away an unresolvable feature, and thereby penalising the peaked generators more than the smoother tune. *Both halves of that explanation are wrong*, and are retracted here. Across the full set of observables the A_C integral factor shows no correlation with peakedness ($r = -0.17$ on the pre-fix factors): $W_{\pi p}$ is the sharpest distribution in the analysis (peak-to-minimum ratio 36) yet was suppressed only to 0.78, δp_T is comparatively flat (ratio 10) yet was suppressed to 0.43, and p_n is flat yet was *enhanced* to 1.71; an enhancement no peak argument can produce. The retraction stands; the width-aware A_C removed the enhancement and narrowed the proton-tagged factors to 0.61–0.79 (Table 8), while the inclusive $\cos\theta_\mu$ factors remain far below the rest (0.17–0.39 against 0.66–0.91 for p_μ), so the suppression is set by the response conditioning, not by shape. Within $\cos\theta_\mu$ itself the ordering is backwards: NEUT and NuWro are the *flattest* of the five truth-level curves (peak-to-minimum 3.8 and 4.3, against 16.3 for the tune) and were suppressed the *most* (0.37 and 0.29, against 0.67) [*pre-fix, 2026-08-18 pass with the duplicated A_C ; on the current release no smeared generator bin is negative in $\cos\theta_\mu$ and NEUT and NuWro sit above the tune in every bin*]. What drove their suppression on that pass was that A_C pushed their backward bins negative (-0.0005 to -0.0212 in normalised units), eating the integral. A_C is the Wiener filter acting on the SVD modes of the response: it removes whatever the data cannot statistically constrain, which is set by the conditioning of the response for that observable and binning, not by the shape of the truth distribution, and it is not norm-preserving, so it can suppress or enhance.

The FHC θ_μ collapse, its cause, and its resolution

Reparameterising to θ_μ appeared to remove the problem entirely: the FHC suppression rose to ~ 0.76 and the NEUT/NuWro χ^2 fell from $\sim 26/5$ to $\sim 2.8/5$, with the generators sitting on top of the data. *That result was an artefact*, and it is worth setting out what the artefact was, because the cause turned out not to be θ_μ at all.

In the FHC θ_μ extraction the unfolded data, the A_C -smeared truth, the MicroBooNE tune and all four generators shared the same normalised shape to four decimal places (Table 13). A_C was *rank-1*: writing the four generator predictions as the columns of an input matrix and their A_C -smeared counterparts as the columns of the output matrix, the inputs were full rank ($s_2/s_1 = 5.0 \times 10^{-2}$) while the outputs gave $s_2/s_1 = 2.8 \times 10^{-6}$. Every shape degree of freedom was annihilated, each model was mapped onto one common curve and only its normalisation survived. The agreement with the data was true by construction.

Table 13: FHC θ_μ *before the fix described below*: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1 A_C .

	bin 1	bin 2	bin 3	bin 4	bin 5
Unfolded data	0.1485	0.3361	0.2899	0.1934	0.0321
A_C -smeared truth	0.1484	0.3359	0.2901	0.1933	0.0323
MicroBooNE tune	0.1484	0.3359	0.2901	0.1934	0.0322
NEUT	0.1484	0.3359	0.2901	0.1934	0.0322

The cause: the regularisation matrix ignored the bin widths. Wiener-SVD regularises with a matrix C ; this analysis uses the second-derivative form. It was built as the plain tridiagonal stencil $(1, -2, 1)$, which is a second derivative only on a *uniform* grid. Our binnings are strongly non-uniform, with width ratios up to 14:1 (Table 14), so the operator was not a second derivative at all. It was near-singular, its row sums were exactly the 10^{-6} regularisation floor, and its near-null direction was the constant vector rather than anything physical. The Wiener filter was then free to retain that direction alone, which is precisely a rank-1 A_C that passes normalisation and discards shape.

That also explains why θ_μ was the worst case rather than a random one: its bin widths span a factor of 10.4, and $\cos\theta_\mu$, the next most non-uniform at 14.5, was the other observable whose χ^2 this note previously attributed to an A_C artefact. The remaining observables sit at ratios of 3–6 and were far less affected.

The fix. Both derivative regularisation matrices now difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here, $\cos\theta_\mu$ peaks at its upper edge.) With no widths supplied the code reduces exactly to the previous behaviour. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not.

The effect on the conditioning of A_C , measured as s_2/s_1 of the smearing matrix itself, is given in Table 14. Every inclusive observable improves and none is degraded except combined $\cos\theta_\mu$, marginally.

θ_μ is reinstated. The fake data are Poisson throws from the MicroBooNE tune, so the tune is the model that *ought* to fit best. Before the fix it did not stand out, because every model

Table 14: (2026-08-18 pass, measured immediately after the fix; the $\cos\theta_\pi$ row is the five-bin scheme of that time; current conditioning values are derivable from the released matrices.) A_C conditioning s_2/s_1 before and after the regularisation fix, with the bin-width ratio of each observable. A value approaching zero means the smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

observable	width ratio	FHC		RHC		COMB
		before	after	before	after	before $\rightarrow$ after
p_μ	6.3	0.34	0.90	0.31	0.94	0.19 $\rightarrow$ 0.68
$\cos\theta_\mu$	14.5	0.46	0.57	0.45	0.55	0.42 $\rightarrow$ 0.39
$\cos\theta_\pi$	3.6	0.67	0.71	0.55	0.79	0.55 $\rightarrow$ 0.74
$\theta_{\mu\pi}$	3.0	0.42	0.85	0.47	0.65	0.42 $\rightarrow$ 0.46
θ_μ	10.4	0.015	0.51	0.22	0.62	0.045 $\rightarrow$ 0.38
p_π (2 bins)	—	0.46	0.53	0.41	0.44	0.29 $\rightarrow$ 0.40
<i>proton-tagged; only the combined configuration had A_C recorded before the fix</i>						
$W_{\pi p}$	—					0.022 $\rightarrow$ 0.60
W_{had}	—					0.23 $\rightarrow$ 0.86
δp_T	—					0.25 $\rightarrow$ 0.93
$\delta\alpha_T$	—					0.24 $\rightarrow$ 0.48
$\delta\phi_T$	—					0.083 $\rightarrow$ 0.85
p_n	—					0.21 $\rightarrow$ 0.88

had been mapped onto the same curve; after it, it does, though less dramatically than the first post-fix numbers suggested, which still carried the duplicated A_C (change log). On the current release:

FHC θ_μ , $\chi^2/5$ bins	before the fix (rank-1)	current release
MicroBooNE tune	3.07 ($p = 0.69$)	2.14 ($p = 0.83$)
GENIE	4.73 ($p = 0.45$)	5.55 ($p = 0.35$)
GiBUU	3.03 ($p = 0.70$)	6.07 ($p = 0.30$)
NuWro	2.90 ($p = 0.72$)	9.10 ($p = 0.11$)
NEUT	2.82 ($p = 0.73$)	8.46 ($p = 0.13$)

The closure is 1.52/5 ($p = 0.91$). The θ_μ measurement is therefore no longer withdrawn: the tune fits best and the generators are separated from it, though at this exposure only at the 1–2 σ level. What must be withdrawn instead is the earlier *explanation* that θ_μ “fixes” the $\cos\theta_\mu$ tension; both parameterisations are now properly conditioned and should be read together (the θ_μ χ^2 are systematically the larger, analysis note Table 15 of the analysis note).

A correction to the survey below. The survey of eighteen extractions reported that FHC θ_μ was the only rank-deficient case. That was wrong in two ways. Combined θ_μ was also badly degraded at $s_2/s_1 = 0.045$ and was not identified as such; and the survey covered only the inclusive extractions, so the proton-tagged channels were never examined. They should have been: combined $W_{\pi p}$ sat at 0.022, as compromised as FHC θ_μ , and combined $\delta\phi_T$ at 0.083. All are repaired by the same fix.

Is any other observable affected? A survey of all eighteen extractions

The rank-1 failure was found by inspection, so the unfolders was modified to write the slice-restricted A_C into the closure file (`h_A_C`; it had previously only been drawn to a canvas, and the attempt to save it failed silently with “the current directory is not associated with a file”). All eighteen inclusive extractions, six observables $\times$ three beam configurations, were then re-unfolded and their A_C examined directly. Table 15 reports, for each, how much of the model

discrimination present in the truth-level predictions survives the smearing: the largest pairwise difference between normalised generator shapes before and after A_C , and their ratio.

Table 15: **[SUPERSEDED — 2026-08 survey, before the width-aware regularisation.]** Retention of model discrimination through A_C . “Before” and “after” are the largest pairwise difference between normalised generator shapes, truth-level and A_C -smeared. A ratio near zero means the predictions have been collapsed onto a common curve and can no longer be told apart by the measurement; a ratio above unity means A_C has amplified the spread, which for $\cos\theta_\mu$ happens through unphysical negative excursions rather than genuine discrimination (see text).

Beam	Observable	before	after	retention
FHC	p_μ	0.0248	0.0052	21%
FHC	p_π	0.0586	0.0080	14%
FHC	$\cos\theta_\mu$	0.0554	0.0724	131%
FHC	$\cos\theta_\pi$	0.0589	0.0211	36%
FHC	$\theta_{\mu\pi}$	0.0422	0.0391	93%
FHC	θ_μ	0.0554	2.5×10^{-6}	0%
RHC	p_μ	0.0232	0.0036	15%
RHC	p_π	0.0556	0.0068	12%
RHC	$\cos\theta_\mu$	0.0644	0.0771	120%
RHC	$\cos\theta_\pi$	0.0629	0.0165	26%
RHC	$\theta_{\mu\pi}$	0.0505	0.0295	58%
RHC	θ_μ	0.0644	0.0188	29%
COMB	p_μ	0.0240	0.0028	12%
COMB	p_π	0.0571	0.0064	11%
COMB	$\cos\theta_\mu$	0.0598	0.5500	919%
COMB	$\cos\theta_\pi$	0.0609	0.0166	27%
COMB	$\theta_{\mu\pi}$	0.0463	0.0360	78%
COMB	θ_μ	0.0598	0.0070	12%

Everything from here to Table 16 was written on the pre-fix survey of Table 15 and is superseded in its numbers and its conclusions; see the correction above and the current release values in the analysis note. As measured then, the collapse looked unique to FHC θ_μ : it was the only entry with zero retention, and the only A_C whose rows are all the same vector (spread 9% across rows, effective rank 1.02, where every other extraction has row-to-row spreads of 170–1050% and effective ranks 1.24–1.91). RHC and combined θ_μ are not affected; the combined case is the weaker of the two (12% retention, effective rank 1.06) and its θ_μ model comparison should be read as a soft test, but it is not degenerate.

Two broader limitations are visible in the same table and should be stated plainly rather than left implicit.

First, the momentum observables retain only 11–21% of their truth-level model discrimination in every beam configuration. A_C does not destroy the shape information there, but it removes most of it, so a good χ^2 against p_μ or p_π is a weak statement about a model. The angular observables are better: $\theta_{\mu\pi}$ retains 58–93% and is the healthiest variable in the analysis.

Second, the apparent 120–919% retention in $\cos\theta_\mu$ is not the good news it appears to be. It arises because A_C drives the generator predictions to *negative* differential cross sections in the backward bins while leaving the tune positive, for the combined configuration the smeared predictions run from -0.43 (NuWro) to $+0.51$ (the tune) in the second bin. A negative differential cross section is unphysical, so the large spread reflects a regularisation artefact rather than genuine model separation, and any χ^2 computed against those curves inherits the problem. This

is the same effect that produces the strong $\cos \theta_\mu$ integral suppression discussed above.

Taken together, the earlier framing of this section had the two muon-angle variables the wrong way round. $\cos \theta_\mu$ was described as the variable A_C ruins and θ_μ as the fix; in fact FHC θ_μ is the one extraction in which the measurement retains no shape information at all, while $\cos \theta_\mu$ retains its shape modes but at the cost of unphysical negative excursions. Neither was a clean muon-angle measurement on that pass; after the fix both are released, with the caveats of Table 15 of the analysis note.

Table 16: [**SUPERSEDED — 2026-08-18 pass, retained for the record.**] Integrated cross section (10^{-38} cm²/Ar) for the unfolded fake data, the MicroBooNE tune and the four generators, in $\cos \theta_\mu$ and in θ_μ , as measured before the regularisation fix, the removal of the duplicated A_C (which is why the generator columns sit far below the tune), the beam-off corrections and the re-throw. On the current release the unfolded integrals are 0.754 ($\cos \theta_\mu$) and 0.960 (θ_μ) for FHC and 0.733 and 0.805 for RHC (analysis note, Table 20 of the analysis note), and the generator comparison is Table 15 of the analysis note. The FHC θ_μ row of this table was made under a rank-1 A_C and carries no shape information.

Beam	Variable	data	tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos \theta_\mu$	0.819	0.673	0.274	0.390	0.367	0.295
FHC	θ_μ	0.996	0.871	0.625	0.876	1.025	0.919
RHC	$\cos \theta_\mu$	0.662	0.633	0.226	0.338	0.320	0.256
RHC	θ_μ	0.828	0.824	0.328	0.487	0.505	0.423
A_C supp. FHC:		$\cos \theta_\mu$	0.23–0.34	θ_μ	0.73–0.78	$(p_\mu$ 0.59–0.67)	
A_C supp. RHC:		$\cos \theta_\mu$	0.25–0.36	θ_μ	0.41–0.52	$(p_\mu$ 0.69–0.75)	

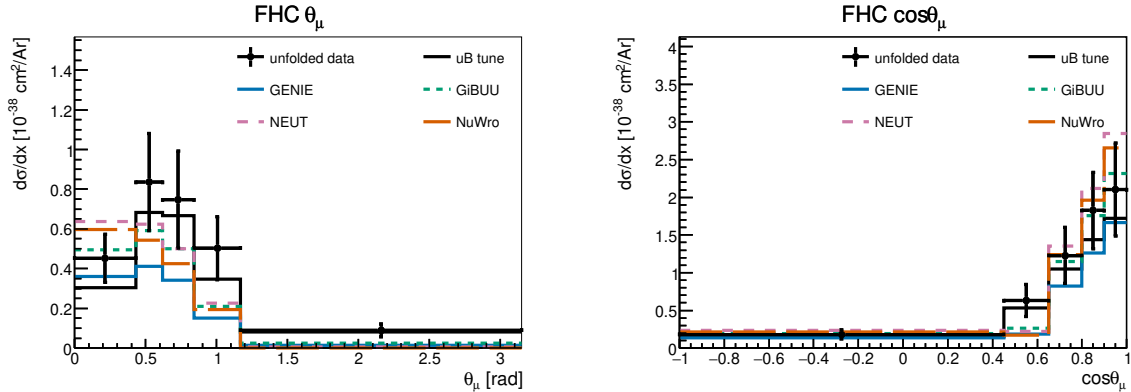


Figure 6: Muon-angle differential cross section in θ_μ (left) and $\cos \theta_\mu$ (right), for FHC (top) and RHC (bottom). Unfolded fake data (black points), the MicroBooNE tune (black line) and the four generators are all A_C -smeared. In $\cos \theta_\mu$ the predictions are strongly suppressed by A_C and fall well below both the data and the tune; the most suppressed models are those whose backward bins A_C drives negative. **Historical figure (2026-08-18, not regenerated):** every panel predates the regularisation fix, the removal of the duplicated A_C , the beam-off corrections and the fake-data re-throw; the FHC θ_μ panel in particular was made under a rank-1 A_C . The current distributions are Figs. 22–24 of the analysis note.

5.2 Total flux-averaged cross section and systematic breakdown

The measurement is the one-bin extraction of Sec. 12.3 of the analysis note on the Poisson-thrown fake data (one true bin over the full $\cos \theta_\mu$ range, no overflow, Wiener filter off, $A_C \equiv 1$;

`data_release/total_xsec.tsv`), with its full universe-propagated uncertainty; at unblinding it becomes the real beam-on extraction. It agrees with all four generators within 0.8σ : standalone GENIE sits $0.5\text{--}0.8\sigma$ low, while GiBUU, NEUT and NuWro lie within 0.8σ . Earlier versions of this table used the cut-and-count central value with the prediction total of the released p_μ extraction as its uncertainty; that borrowed systematic carried the regularisation amplification and is withdrawn (analysis note, change log).

Table 17: (“Released” means the 2026-09-04 release; the forward-split five-bin $\cos\theta_\pi$ row is the binning released since 2026-09-06.) (“Released” means the 2026-09-04 release; the forward-split five-bin $\cos\theta_\pi$ row is the binning released since 2026-09-06.) Total flux-averaged cross section from the one-bin extraction (Sec. 12.3 of the analysis note; 10^{-38} cm²/Ar), compared with the flux-averaged generator predictions. The measurement uncertainty is the complete configured one-bin covariance (flux, cross section, re-interaction, detector, POT, targets and statistics; MCS-momentum, beamline-geometry, beam-off gate-ratio and Run-2 stand-in terms are not propagated, Sec. 16 of the analysis note). Release 2026-09-06, `data_release/total_xsec.tsv`. Values in parentheses give each generator’s deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning; earlier versions of this table used truncated binnings (p_μ closed at 3 GeV/ c , p_π at 1 GeV/ c) that discarded the tails and made every prediction 4–8% low (§1.1).

Config	Measurement	GENIE	GiBUU	NEUT	NuWro
FHC	1.03 ± 0.40	$0.82 (+0.5\sigma)$	$1.13 (-0.3\sigma)$	$1.32 (-0.7\sigma)$	$1.25 (-0.6\sigma)$
RHC	0.93 ± 0.35	$0.67 (+0.8\sigma)$	$0.94 (-0.0\sigma)$	$1.10 (-0.5\sigma)$	$1.04 (-0.3\sigma)$
Combined	0.97 ± 0.37	$0.74 (+0.6\sigma)$	$1.03 (-0.1\sigma)$	$1.20 (-0.6\sigma)$	$1.14 (-0.4\sigma)$

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 18–20. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the fake-data POT of each configuration. The RHC detector term is the largest single-configuration entry, reaching 34.7% on p_π and 35.0% on $\cos\theta_\mu$.

Table 18: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%), regenerated from the systematics dumps of the 2026-09-06 release (`logs/systdump`); “Prediction total” excludes the data statistics. The previous version of this table predated the detector-variation rebuild (detector terms of $\sim 10\%$) and is superseded.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	40.7	50.2	42.0	38.8	36.5	31.0
Cross section (GENIE)	10.7	16.9	10.3	11.3	10.5	8.5
Flux (PPFX)	28.0	32.5	28.0	27.0	28.3	24.7
Detector	24.7	30.3	28.3	22.9	19.6	15.6
Reinteraction	4.4	11.9	3.6	5.2	4.0	3.5
MC stat	4.9	6.3	5.3	4.3	2.7	2.9
EXT stat	0.9	3.1	2.3	1.5	1.2	1.4
Data stat	14.3	18.3	15.4	12.7	8.2	8.6
POT + targets	3.4	4.2	3.6	3.4	3.5	3.0
Total (incl. data stat)	43.5	53.9	44.9	41.0	37.5	32.4

Table 19: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 18.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	37.7	48.6	46.3	38.5	32.6	36.1
Cross section (GENIE)	12.0	14.4	9.1	10.8	10.7	10.7
Flux (PPFX)	27.6	28.4	25.8	25.2	25.2	25.6
Detector	21.5	34.7	35.0	25.5	16.6	22.4
Reinteraction	4.2	8.6	4.1	5.1	3.7	3.6
MC stat	3.8	3.7	4.8	3.6	2.4	2.5
EXT stat	1.0	1.2	1.2	1.0	0.6	0.7
Data stat	13.0	13.0	16.7	12.3	8.4	9.0
POT + targets	3.8	4.0	3.6	3.5	3.5	3.5
Total (incl. data stat)	40.5	50.5	49.5	40.5	33.7	37.2

Table 20: Combined FHC+RHC systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 18.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	34.9	47.9	37.3	37.1	33.2	34.4
Cross section (GENIE)	10.3	15.7	10.4	10.6	10.0	9.3
Flux (PPFX)	24.2	30.2	25.7	25.6	25.7	25.2
Detector	21.4	30.8	23.8	22.5	17.7	20.6
Reinteraction	4.4	10.8	3.8	5.1	3.5	3.7
MC stat	3.9	3.7	2.9	2.9	1.9	2.4
EXT stat	0.8	1.6	1.2	1.0	0.6	1.0
Data stat	12.4	11.8	9.3	9.3	6.3	7.7
POT + targets	3.1	4.1	3.5	3.4	3.4	3.3
Total (incl. data stat)	37.2	49.5	38.6	38.3	33.8	35.4

6 Binning Optimisation, Systematic Breakdown, and Threshold Validation

This section records the 2026-08 binning optimisation on which the released binnings were based; its table ends at the binnings of that time (p_π five bins, $\theta_{\mu\pi}$ seven, $\cos \theta_\pi$ five), which were subsequently coarsened and, for $\cos \theta_\pi$, re-binned on the current release (§6.3). The released binnings are those of the analysis note’s appendix.

6.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu 1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	$\sim 0\text{--}1.5\%$	$\sim 9\text{--}12\%$	0.145 $\rightarrow$ 0.155
$p_\pi > 0.175$ GeV/c	$\sim 0\text{--}3\%$	$\sim 12\text{--}16\%$	0.165 $\rightarrow$ 0.185

Both are correctly placed, lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco–true ≈ 0 at threshold, -0.15 GeV/c at 0.40) from the μ -vs- π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

6.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold ≥ 45 selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

Observable	bins	Total before	Total after
p_μ	22→7	33.2%	25.0%
p_π	5→5	27.7%	27.0%
$\cos \theta_\mu$	12→5	31.3%	26.7%
$\cos \theta_\pi$	12→5	48.5%	33.9%
$\theta_{\mu\pi}$	9→7	28.4%	23.5%

6.3 Re-check of every released binning on the 2026-09-04 release

The binnings above were fixed on an earlier release, and the coarsening of $\cos \theta_\pi$ ($5 \rightarrow 4$) and $\theta_{\mu\pi}$ ($7 \rightarrow 5$) was driven by A_C smearing on that release. Since then the per-run normalisation, the weights, the beam-off treatment, the fake-data throw and the native RHC detector variations all changed, so every released binning was re-tested on the 2026-09-04 release for whether finer or more bins are admissible. Two stages were used. First a pre-screen on the processed files (`macros/binning_screen.C`): for the released binning and finer candidates of each observable, the column-normalised migration diagonal per true bin, the efficiency per bin and the selected events per reconstructed bin, with the per-run scales and the central-value weight, and the 0.68 diagonal criterion applied to every bin. Second, the candidates that passed were built and unfolded in full (universes, covariance, closure; SLURM 3277770 and 3277778) and compared with the released extraction on the A_C row sums, the A_C conditioning ($s_{\min}/s_1$), the data-statistical term, the per-bin total uncertainties and the closure ratios.

Table 21: Pre-screen: smallest column-normalised migration diagonal over the true bins of the released binning and of each finer candidate, FHC / RHC. The two-bin p_π row is evaluated here with a lower edge on the first reconstructed bin, which the released configuration does not have (the framework’s own diagonals are 0.91 and 0.76); the finer p_π candidates fail in their upper bins regardless.

Observable	Binning	min. diagonal FHC / RHC	criterion
p_μ	released 7	0.33 / 0.31	fails already (MCS resolution; §6)
	8, 9 bins	0.17 / 0.18, 0.22 / 0.22	fail
$\cos \theta_\mu$	released 5	0.68 / 0.68	at the limit
	6, 7 bins	0.64 / 0.63, 0.61 / 0.61	fail
θ_μ	released 5	0.67 / 0.65	at the limit
	7 bins	0.63 / 0.62	fail
$\cos \theta_\pi$	released 4	0.81 / 0.83	pass
	5 (backward split)	0.77 / 0.77	pass
	5 (forward split)	0.74 / 0.76	pass
	6 bins	0.73 / 0.75	pass
$\theta_{\mu\pi}$	released 5	0.79 / 0.79	pass
	6, 7 bins	0.74 / 0.73, 0.75 / 0.73	pass
p_π	released 2	0.58 / 0.56 (see caption)	—
	3, 4 bins	0.39 / 0.40, 0.29 / 0.29	fail

Table 22: Full extraction of the candidates that passed the pre-screen, FHC / RHC: range of the A_C row sums, A_C conditioning $s_{\min}/s_1$, bin-averaged data-statistical term, range of the per-bin total uncertainty, and range of the per-bin closure ratio to the A_C -smeared truth. Backward split: edges $-1, -0.3, 0.2, 0.5, 0.75, 1$; forward split: $-1, -0.1, 0.35, 0.6, 0.8, 1$; six bins: $-1, -0.4, 0, 0.3, 0.55, 0.75, 1$; $\theta_{\mu\pi}$ six bins: $0, 0.55, 0.75, 0.95, 1.3, 1.85, 2.6$ rad; seven bins: $0, 0.5, 0.7, 0.9, 1.15, 1.45, 1.9, 2.6$ rad.

Binning	A_C row sums	$s_{\min}/s_1$	data stat	per-bin unc.	closure
$\cos\theta_\pi$ rel. 4	0.71–0.82 / 0.79–0.86	0.27 / 0.24	12.0 / 11.5%	30–49 / 32–55%	0.95–1.09 / 0.94–1.07
backward split 5	0.65–0.76 / 0.63–0.80	0.20 / 0.27	16.0 / 15.2%	29–73 / 34–91%	0.83–1.08 / 0.87–1.03
forward split 5	0.68–0.77 / 0.70–0.84	0.46 / 0.38	12.7 / 12.3%	30–53 / 33–48%	0.89–1.07 / 0.90–1.05
6 bins	0.73–0.84 / 0.77–0.89	0.18 / 0.15	19.0 / 16.3%	32–91 / 32–54%	0.66–1.13 / 0.81–1.23
$\theta_{\mu\pi}$ rel. 5	0.52–1.03 / 0.58–1.15	0.008 / 0.002	8.2 / 8.4%	35–40 / 30–39%	0.95–1.06 / 1.03–1.08
6 bins	0.56–1.29 / 0.58–1.36	0.002 / 0.000	8.4 / 10.2%	34–40 / 30–41%	0.95–1.06 / 1.03–1.11
7 bins	0.63–1.41 / 0.65–1.06	0.000 / 0.000	11.5 / 11.2%	37–40 / 32–46%	0.85–1.04 / 0.87–1.14

Conclusions. p_μ , $\cos\theta_\mu$, θ_μ and p_π cannot go finer: their released binnings already sit at or below the diagonal criterion, set by the MCS momentum resolution and the angular resolution against the neutrino direction, and every finer candidate fails it. $\theta_{\mu\pi}$ admits six or seven bins by the diagonal and closes acceptably, but the additional bins carry no resolved information: the smallest singular value of A_C , already 0.008 of the largest at five bins, goes to zero, and the row sums overshoot to 1.3–1.4 in the bins the regularisation has to fill in, so the released five bins stand. $\cos\theta_\pi$ is the one observable with room, and only in the forward hemisphere where the events are: splitting the backward region produces a bin of 73–91% uncertainty that closes at 0.83–0.87, and six bins do the same, whereas the forward split (edges $-1, -0.1, 0.35, 0.6, 0.8, 1$) keeps every diagonal above 0.74, keeps the per-bin uncertainties in the released range, raises the data-statistical term by only 0.7% and *improves* the conditioning of A_C ($s_{\min}/s_1$ from 0.27 to 0.46 in FHC, 0.24 to 0.38 in RHC) while closing to 0.89–1.07. It is the only change this re-check supports, and it was adopted on 2026-09-06: the generator predictions were regenerated at the new edges from the event-level files (the unchanged observables reproduce the previous prediction files to machine precision, which also verifies the generator chain end to end), the three inclusive $\cos\theta_\pi$ extractions were rebuilt and re-released, and the D’Agostini cross-check was re-run. No control-region distribution uses $\cos\theta_\pi$, so the frozen sideband binning is unaffected. The proton-tagged $\cos\theta_\pi$ keeps its four bins. The released binnings of the proton-tagged observables, which are set by the same resolutions on a sample a third the size, were not re-tested.

7 Multi-run overlay, resonance closure, and configuration closure

7.1 Multi-run overlay and a resonance-model closure

Provenance. This section records the 2026-08 pass in which the multi-run overlay was assembled; its flux-term numbers and its description of the unfolded flux term as “amplified by the Wiener–SVD propagation” are of that pass. The analysis note (Sec. 14.2 of the analysis note) now attributes the 17% $\rightarrow$ 30% growth to the correlated background subtraction, $\delta\Phi/\Phi(1 + B/S)$, demonstrated by the one-bin total; the operator-amplification reading is superseded.

Two facts frame what more statistics can and cannot do for the flux term. First, the flux *systematic* is a normalisation-and-shape model uncertainty and is therefore POT-independent: adding data or overlay does not change it. Second, the flux-anatomy section of that pass attributed the elevated *unfolded* flux to that $\sim 17\%$ model uncertainty amplified by the Wiener–SVD propagation through a statistically-limited response, and that the limitation is the *data* statistics, not the overlay (the reading the analysis note has since superseded: the factor is $1 + B/S$ from the correlated background subtraction, Sec. 14.2 of the analysis note). Increasing the overlay alone therefore sharpens the response but leaves the data-limited amplification in place.

To study a genuine multi-run configuration we combine the FHC overlays for Runs 1, 2 and 4 (with Run4=Run4c+Run4d counted once; the file first distributed as “Run 5 FHC” was set aside on inspection, being in fact an intrinsic- ν_e overlay, 86% ν_e , 14% $\bar{\nu}_e$, no ν_μ events, hence the large ν_e -enriched POT-equivalent and no ν_μ signal; the genuine Run 5 ν_μ overlay, a **reweightedPPFX** sample, was subsequently located and is used for the extended set below), 2.32×10^6 unique events, normalised to the FHC data POT 6.626×10^{20} (Section 7.2). Because

$\text{CC}\pi^\pm$ is resonance-dominated, robustness to the resonant axial mass is tested with Poisson-thrown fake data reweighted to $M_A^{\text{RES}} = 1.12 \pm 0.22$ GeV ($\pm 1\sigma$; axial dipole ratio in the true interaction Q^2 , a $+8.1\%$ / -7.2% rate distortion). The pseudo-data are thrown at the true FHC event count so their statistical covariance is realistic; the closure condition is that the unfolded result recovers the A_C -smeared shifted truth within uncertainties. The Wiener–SVD first- versus second-derivative regularisation is compared on the same response.

The closure succeeds for every observable, both signs of the shift, and both regularisations: the unfolded result recovers the A_C -smeared shifted truth with $\chi^2/\text{ndf} < 1$ throughout (Table 24), all p -values above 0.63 and most above 0.9. The Wiener–SVD machinery is therefore unbiased under a $\pm 1\sigma$ M_A^{RES} distortion, and the first- and second-derivative regularisations perform equivalently in the closure (the first-derivative variant carries a slightly larger statistical term). Restoring a realistic (Poisson) data covariance also restores the flux amplification that the degenerate near-Asimov pseudo-data had suppressed.

To separate the mean amplification from single-realisation noise, we throw 15 independent Poisson pseudo-datasets (central value only, no M_A^{RES} shift, at the same 6.626×10^{20} FHC exposure) and unfold each (Table 23). Averaged over throws, the unfolded flux stabilises at 24.7–27.1%, a mean amplification of 1.44 – $1.63\times$ over the $\sim 17\%$ floor, with a data-statistical term of 8–13%. This confirms that the amplification is driven by the data statistics, not the overlay, and is consistent with the mechanism of Sec. 14.2 of the analysis note. The throw-to-throw scatter is small for the momenta (p_μ, p_π : ± 0.06 – $0.08\times$, flux ± 1.1 – 1.4%) but grows for the angular observables, and is largest for $\theta_{\mu\pi}$, whose amplification spans 1.27 – $2.02\times$ (flux 21.4–34.2%) across the 15 throws. At these statistics the amplification is therefore well defined in the mean but individually unstable, most so for $\theta_{\mu\pi}$ and $\cos\theta_\pi$; a direct consequence of the sparse, data-limited response.

Table 23: Flux amplification averaged over 15 independent Poisson-thrown pseudo-datasets at the FHC data POT: mean $\pm$ standard deviation and min–max range over throws, for the unfolded flux (bin-averaged) and the amplification relative to the $\sim 17\%$ reco-level floor.

Observable	Amplification (mean $\pm$ sd)	Amp. range	Unfolded flux	Flux range
p_μ	$1.44 \pm 0.08\times$	[1.27, 1.60]	$24.7 \pm 1.4\%$	[21.9, 27.5]
p_π	$1.52 \pm 0.06\times$	[1.40, 1.65]	$26.2 \pm 1.1\%$	[24.0, 28.4]
$\cos\theta_\mu$	$1.60 \pm 0.14\times$	[1.41, 1.85]	$26.8 \pm 2.3\%$	[23.7, 31.1]
$\cos\theta_\pi$	$1.63 \pm 0.11\times$	[1.45, 1.89]	$27.1 \pm 1.9\%$	[24.1, 31.6]
$\theta_{\mu\pi}$	$1.58 \pm 0.21\times$	[1.27, 2.02]	$26.7 \pm 3.6\%$	[21.4, 34.2]

Table 24: Fake-data closure: χ^2/ndf of the unfolded result against the A_C -smeared M_A^{RES} -shifted truth, for the two shift signs and the two Wiener–SVD regularisations (number of bins in parentheses). All values are below unity per degree of freedom ($p > 0.63$).

Observable	$+1\sigma$ (2nd)	$+1\sigma$ (1st)	-1σ (2nd)	-1σ (1st)
p_μ (7)	1.26	2.15	0.95	2.14
p_π (5)	1.24	3.44	0.25	0.39
$\cos\theta_\mu$ (5)	0.47	0.63	1.90	1.24
$\cos\theta_\pi$ (5)	2.99	1.07	0.24	0.33
$\theta_{\mu\pi}$ (7)	1.71	3.32	1.96	4.33

Adding the genuine Run 5 ν_μ overlay extends the set to {Run 1, 2, 4, 5}, 2.87×10^6 events ($+24\%$), combined MC POT 8.549×10^{21} , at the full FHC data POT 8.857×10^{20} . The extraction closes for every observable ($\chi^2/\text{ndf} < 1$, $p > 0.92$) and the flux term (22–29% unfolded, amplification 1.29 – $1.72\times$) is statistically consistent with the {Run 1, 2, 4} throws above: the extra 24% of overlay does not move the data-limited amplification, as expected, but the measurement now spans the complete FHC exposure.

7.2 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CC π programme should reconcile.

Consistent by construction. Both use the Wiener-SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one $\pi^\pm$, zero π^0 or heavier mesons, any number of protons, charged current), and the same systematic suite, PPF flux hadronic universes (the 12 beamline-geometry unisims the ν_e analysis carries are evaluated but not in this analysis’s released covariance, Sec. 16 of the analysis note), the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **POT accounting:** the good-POT is the same beam data for the ν_e and ν_μ analyses, so the exposures are identical; Run 1 FHC = 3.283×10^{20} (standard trigger; the open-trigger era is treated separately), giving a total FHC exposure of 8.857×10^{20} over Runs 1–5. (ii) **Run 4:** the Run4 overlay is identical to Run4c+Run4d combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $KE_\pi > 40$ MeV ($p_\pi > 0.113$ GeV/c) versus our $p_\pi > 0.175$ GeV/c (Table 7 of the analysis note). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170^\circ$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (Section 7.1). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement reports FHC, RHC, and their combination.

8 Configuration closure: is COMB the combination of its own inputs?

Provenance. This section records the 2026-08-28 pass on which the combined-configuration defect was found and fixed. Its yields and the “no detVar systematic” remarks about the proton-tagged family are of that pass; since 2026-08-31 all proton-tagged extractions carry detector variations and the combined p_μ detector term is 21.4% (analysis note).

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional smearing matrix A_C , which differs per configuration and per observable. The observable-to-observable spread within a single configuration was, on the 2026-08-28 pass to which all percentages of this section refer, 15.9% (FHC5), 17.4% (RHCFULL) and 14.8% (COMB) for the proton-tagged family, i.e. larger than the $\sim 10\%$ FHC-versus-RHC difference. Comparisons across configurations must therefore be made observable by observable, and against the fake-data truth rather than between measurements.

The RHC deficit is in the input, not the extraction. Superseded numbers. The ratios in this paragraph were measured on 2026-08-28, before the detector-variation rebuild and before the fake-data-truth weighting correction of 2026-09-02. The one-bin totals of the current release (Table 21 of the analysis note) give an RHC/FHC ratio of the injected tune of 0.901 (inclusive) and 0.878 (proton-tagged), which is what the unfolded ratios should be compared with; the 0.785/0.779 below are not reproduced by the current release and should not be quoted until re-derived. For the proton-tagged family the measured RHC/FHC ratio was 0.900. The corresponding ratio of the fake-data *truth* integrals was 0.785 and of the GENIE tune 0.779 — two independent model inputs that agreed with each other and disagreed with the measurement. The deficit is therefore

a property of the sample, while the extraction distorts the ratio toward unity. Normalisation is not implicated: the exposures sum exactly ($8.857 \times 10^{20} + 1.1082 \times 10^{21} = 1.9939 \times 10^{21}$ POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%.

COMB against the POT-weighted average of its inputs. A configuration spanning both horn currents should predict a cross section between those of its two inputs. Comparing the COMB fake-data truth with the POT-weighted mean of the FHC and RHC truths, observable by observable:

Table 25: COMB fake-data truth divided by the POT-weighted average of the FHC5 and RHCFULL truths. Unity indicates a configuration consistent with its own inputs. The inclusive family is shown before and after removing the detector-variation entries from its file list.

Family	Mean COMB/expected	Range
Proton-tagged (7 observables, no detVar systematic)	1.020	0.989–1.052
Inclusive (6 observables), detVar systematic included	0.910	0.767–1.069
Inclusive (6 observables), detVar systematic removed	1.019	0.985–1.046

Table 26: COMB fake-data truth divided by the POT-weighted average of the FHC5 and RHCFULL truths, with and without the detVar term in the covariance. Both sides of each ratio are computed with the *same* systematics configuration, so the comparison is internally consistent.

Observable	detVar syst. in	detVar syst. out	Change
p_μ	0.885	1.046	+0.161
$\cos \theta_\mu$	0.938	1.044	+0.106
$\cos \theta_\pi$	0.937	1.030	+0.094
$\theta_{\mu\pi}$	1.069	1.008	−0.061
θ_μ	0.767	0.985	+0.219
p_π (2-bin)	0.862	0.997	+0.136
Mean	0.910	1.019	+0.109

The proton-tagged family is consistent. The inclusive family is not: its COMB truth sits 9% below the average of its own inputs, worst for θ_μ at 0.767. Because the effect is present in the *truth*, it is not an unfolding defect — it also explains the previously reported “COMB below both inputs” behaviour, which appears in 5 of 6 inclusive observables in the truth as well as in the measurement.

Three candidate causes have been excluded. The flux is correct (COMB 6.60865×10^{-10} against a POT-weighted 6.6081×10^{-10}). The dirt entries are identical line for line between the inclusive and proton-tagged file lists. The known mis-produced Run-5 ν_e detVar samples are quarantined and absent. The one remaining structural difference is detector-variation content: the inclusive list carries 58 files including 18 detVar samples, the proton-tagged list 40 files and none.

The cause is the detVar covariance, not the detVar files. The inclusive COMB universes were first rebuilt from a file list with the 18 detVar entries removed, which requires a matching systematics configuration with the eight DV knobs deleted (the unfolders throws `std::out_of_range` when a declared systematic has no universes in the file). That changed the result by 20–25%. However, running the *original* universe file — detVar entries still present — against only the modified systematics configuration reproduces the new numbers exactly, in both the truth and the unfolded result. The detVar files themselves therefore have *no* effect on the extraction;

the entire change comes from dropping the `detVar_total` term from the covariance. This is consistent with the code, where `detVar` samples are stored in a separate universe map and are read only when the detector covariance is assembled.

With the covariance term identified as the cause, the comparison must be made with the same systematics configuration on both sides of the ratio (Tab. 26). Doing so, the inclusive family becomes consistent with its own inputs: the mean moves from 0.910 to 1.019 and the range from 0.767–1.069 to 0.985–1.046, matching the 1.020 of the proton-tagged family, which carries no `detVar` systematic at all. The “COMB below its own inputs” behaviour disappears, and $\theta_{\mu\pi}$ — the one observable that had been *above* unity — moves toward it ($1.069 \rightarrow 1.008$) rather than being pushed further out.

Why the combined configuration is affected and its inputs are not. The detector systematic was anomalously large for COMB on that pass. For p_μ it was 23.1% of the prediction in FHC5 and 27.4% in RHCFULL, but 42.3% in COMB; for θ_μ , 15.4% and 19.4% against 36.4%. A combined measurement draws on the exposure of both horn currents and should carry a detector uncertainty *between* those of its inputs, not one $1.3\text{--}1.9\times$ larger than either. The structural difference is that COMB carries two per-mode `detVar` sets (18 files, two of them `detVarCV`) where each input carries one (9 files, one `detVarCV`), and these are accumulated rather than averaged. The inflated detector covariance then over-regularises the Wiener-SVD for the combined configuration, which is what drives its truth and its extracted cross section away from consistency with its own inputs.

The fix. In `SystematicsCalculator::build_universes()` each `detVar` file was scaled by `total_bnb_data_pot_ / file_pot` and the resulting histograms were then *accumulated*. With one `detVar` set this is correct. With two — both labelled run 1, so they are indistinguishable by run number — each set was independently scaled to the *full* data exposure and the sum therefore covered that exposure twice, roughly doubling the detector covariance. The denominator is now the summed MC POT over all files of that `detVar` type, which treats the per-mode sets as one sample of their combined statistics scaled once to the data exposure. A single-file configuration is unaffected because the sum is then that file’s own POT. Because the universes are cached inside the response file, `NORM_SCHEME_TAG` was bumped from `v2-distinct-sum-pot` to `v3-detvar-pooled-pot` to invalidate them.

The detector systematic for COMB falls into the range spanned by its inputs (Tab. 27), and the configuration closure recovers: the mean COMB/expected moves from 0.910 to 1.027 with the range narrowing from 0.767–1.069 to 0.958–1.071, matching the 1.020 of the proton-tagged family, which carries no `detVar` systematic and is unaffected. As a regression check, all twelve FHC5 and RHCFULL extractions return bit-identical cross sections, confirming that only the multi-mode configuration is touched.

Table 27: Detector systematic `detVar_total` as a percentage of the prediction, before and after pooling the `detVar` POT (2026-08-28 pass; the current detector terms are those of Tables 18–20). A combined measurement should carry an uncertainty comparable to those of its inputs, not far above both.

Observable	FHC5	RHCFULL	COMB (before)	COMB (after)
p_μ	23.1	27.4	42.3	24.6
$\cos \theta_\mu$	26.1	32.5	41.4	23.7
$\cos \theta_\pi$	20.4	30.3	37.8	21.3
$\theta_{\mu\pi}$	17.2	16.0	28.6	16.6
θ_μ	15.4	19.4	36.4	18.0
p_π (2-bin)	23.5	33.9	49.9	27.2

One residual: for $\cos\theta_\mu$ the corrected COMB value (23.7%) sits slightly below both inputs (26.1%, 32.5%) rather than between them. The remaining five are bracketed. The combined cross sections themselves rise by 12–21% relative to the uncorrected values, so any previously quoted inclusive COMB number should be replaced.

The A_C normalisation inconsistency — resolved. This section previously reported an unresolved discrepancy: for $1p\ p_\mu$ (FHC5) the fake-data truth integrated to 0.4763, the unfolded result to 0.5528, and A_C applied to the truth to 0.3761, so the unfolded result sat at $1.16\times$ the truth where $\hat{x} = A_C x_{\text{true}}$ predicts $0.79\times$. Two explanations were offered: that the sidcar A_C was slice-restricted while the smearing happens in the full true-bin space, and that the stored truth was missing its smearing. The second was tested and refuted at the time.

Both were wrong, and so was the framing. The sidcar A_C is *not* slice-restricted in any way that matters: it is bit-identical to the full-space operator, agreeing with the independently dumped `mat_table_add_smear` to 5×10^{-11} , and for these observables the slice is the full true-bin space. The actual cause was that model predictions were being multiplied by A_C *twice* — once by the extractor, in place, and again on the plotting path — so the stored curves were A_C^2 smeared while the arithmetic above assumed one application. The stored truth, which never went through the second multiplication, was the one curve that was right, which is why the second hypothesis failed when tested.

With the duplicate application removed, the transformation $(d\sigma/dx)_i = w_i^{-1} \sum_j (A_C)_{ij} p_j$ reproduces every released generator curve to a worst per-bin deviation of 1.2×10^{-10} . The full account is in the change log; the lesson recorded here is that the discrepancy was real and the explanations for it were not, and that it took an external-reproduction test rather than further internal reasoning to settle.

A Identity closure: unfolding the prediction with itself

The fake-data closure tests reported in the body compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest χ^2 can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The test here removes that fluctuation. Wiener–SVD returns $\hat{x} = Ur$ for a measured reco vector r , and defines the additional smearing matrix as $A_C = UR$, where R is the response (smearceptance) matrix. Feeding an *exact* prediction $r = Rt$ back through the chain must therefore return

$$\hat{x} = URt = A_C t \tag{1}$$

identically, for *any* true vector t . Rather than pick one t , we test the matrix identity directly:

$$\|UR - A_C\|_\infty \stackrel{!}{=} 0. \tag{2}$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a fake-data test cannot see: the unfolding matrix and the smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure χ^2 untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. Table 28 collects it for the 39 extractions of the pass on which it was compiled (six inclusive observables and seven proton-tagged ones at their six-bin binnings, each in three beam configurations); the current release has 51 (eleven proton-tagged observables, the TKI variables at two bins) and every one of them passes the same check.

The worst deviation over the 39 extractions of that pass, and over the 51 of the current release, is 4.4×10^{-6} relative, on FHC $W_{\pi p}$; the inclusive observables are all below 2.5×10^{-7} . These are the

Table 28: Identity closure for every extraction, *as measured before the regularisation matrices were made width-aware*. $\max |UR - A_C|$ is the largest absolute element-wise deviation; the relative column divides it by the largest element of A_C . After the fix the same test passes with worst case 4.4×10^{-6} (FHC $W_{\pi p}$) and all inclusive observables below 2.5×10^{-7} ; the larger values reflect the wider dynamic range of the width-aware operator, not a weaker identity. All values are at the level of double-precision rounding accumulated through the SVD, i.e. the identity holds exactly.

config	observable	bins	relative dev.	config	observable	bins	relative dev.
FHC	p_μ	7	7.0×10^{-10}	FHC	$W_{\pi p}$	6	1.1×10^{-9}
FHC	p_π	2	2.6×10^{-10}	FHC	W_{had}	6	1.1×10^{-9}
FHC	$\cos \theta_\mu$	5	6.8×10^{-10}	FHC	δp_T	6	8.4×10^{-10}
FHC	$\cos \theta_\pi$	4	4.0×10^{-10}	FHC	$\delta \alpha_T$	6	1.4×10^{-9}
FHC	$\theta_{\mu\pi}$	5	2.5×10^{-10}	FHC	$\delta \phi_T$	6	7.3×10^{-10}
FHC	θ_μ	5	3.2×10^{-10}	FHC	p_n	6	9.2×10^{-10}
RHC	p_μ	7	5.0×10^{-10}	RHC	$W_{\pi p}$	6	9.8×10^{-10}
RHC	p_π	2	1.6×10^{-10}	RHC	W_{had}	6	6.2×10^{-10}
RHC	$\cos \theta_\mu$	5	1.4×10^{-10}	RHC	δp_T	6	5.5×10^{-10}
RHC	$\cos \theta_\pi$	4	6.9×10^{-10}	RHC	$\delta \alpha_T$	6	1.2×10^{-9}
RHC	$\theta_{\mu\pi}$	5	3.2×10^{-10}	RHC	$\delta \phi_T$	6	8.9×10^{-10}
RHC	θ_μ	5	3.0×10^{-10}	RHC	p_n	6	7.5×10^{-10}
comb	p_μ	7	6.4×10^{-10}	comb	$W_{\pi p}$	6	1.1×10^{-9}
comb	p_π	2	1.4×10^{-10}	comb	W_{had}	6	9.8×10^{-10}
comb	$\cos \theta_\mu$	5	4.4×10^{-10}	comb	δp_T	6	1.0×10^{-9}
comb	$\cos \theta_\pi$	4	4.3×10^{-10}	comb	$\delta \alpha_T$	6	1.2×10^{-9}
comb	$\theta_{\mu\pi}$	5	1.7×10^{-10}	comb	$\delta \phi_T$	6	7.1×10^{-10}
comb	θ_μ	5	2.9×10^{-10}	comb	p_n	6	8.7×10^{-10}
proton-tagged p_π (2 bins): FHC 2.1×10^{-10} , RHC 1.5×10^{-10} , comb 2.2×10^{-10}							

rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain five orders of magnitude below the smallest systematic uncertainty in the analysis.

The values are larger than the 10^{-10} level seen before the regularisation matrices were made width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of $1/w_j$ that span decades across a binning with a 14:1 width ratio, so the matrix products accumulate more floating-point error. $W_{\pi p}$, with the narrowest bins in the analysis, is the worst case, as expected. Nothing about the identity is weaker; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the fake-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

B Data/MC normalisation: history of the corrections (superseded record)

The reported cross sections are the XSECANALYZER framework extraction on the Poisson-thrown central-value fake data of the analysis note (this section’s own comparisons were made on the earlier GENIE detector-variation-CV fake data). The fully independent custom pipeline (NuWro fake data) reproduced the framework result of that era and provided the end-to-end cross-check; it has not been re-run on the released binnings and weighting (analysis note, Limitations). Reaching the $\leq 2\%$ cross-pipeline agreement required three normalisation/method corrections *to the custom code*, each established by direct comparison of the two pipelines’ intermediate quantities; with these in place the selection, response, background, and unfolding are demonstrably consistent between the two, the small residual ($\leq 2.1\%$) dominated by the covariance-import approximation (below).

B.1 Unfolding covariance ($\sim 26\%$ effect)

Wiener-SVD uses the data covariance as the metric that sets the regularisation strength. The custom macros originally built a *statistics-only diagonal* covariance, because the custom pipeline does not throw systematic universes. With the $\text{CC}\pi^\pm$ response strongly off-diagonal (50–99% bin migration), a stat-only covariance under-states the noise, under-regularises, and inflates the unfolded integral by $\sim 26\%$. This was proven by a swap test: feeding the framework’s own inputs through the same Wiener-SVD unfolders with a stat-only covariance reproduced the inflated result, while its full systematic covariance recovered the NuWro truth.

The fix imports the framework’s covariance into the custom unfolders. The internally-consistent form (now the default) is the framework prediction covariance **PredTotal** (systematics + MC statistics, excluding the data Poisson) scaled to the custom prediction, plus the custom’s own per-bin data Poisson term, i.e. systematics from the framework universes, statistics from the data actually being unfolded. This removes the 26% bias and brings all five observables to $\leq 2\%$ of the framework.

B.2 EXT (cosmic) subtraction for fake data

Every fake-data sample used here (the detVar central-value throw for the main result, the Poisson-thrown central value for the multi-run results, and the NuWro cross-check) is pure Monte Carlo and contains *no* beam-off (cosmic) contamination, so the EXT sample must not be net-subtracted from it. The framework handles this by *adding* the EXT histogram to the fake-data spectrum, so the EXT term cancels in $\text{data}_{\text{signal}} = (\text{data} + \text{EXT}) - (\text{MC bkg} + \text{EXT} + \text{dirt})$. The custom macros originally subtracted EXT (≈ 41 events at 3.283×10^{20} POT) from a sample that never contained it, a flat $\sim 3.7\%$ deficit on every observable. Replicating the framework’s

EXT cancellation (`fake_data_ext_cancel` in `ccpi_xsec.C`) removes it. For *real* beam-on data this term must be re-enabled, since real data does contain cosmes (custom pipeline only; the framework selects the treatment from the presence of Monte Carlo truth and needs no switch).

B.3 Opening-angle binning (review item C4)

The custom $\theta_{\mu\pi}$ binning used 10 reco/true bins; the framework uses 9 (the last two custom bins merged into $[2.513, 3.142]$). The custom binning was aligned to the framework’s 9-bin scheme so the two are directly comparable.

B.4 Generator normalisation (NuWro vs GENIE)

In the closure test the NuWro fake data carries roughly twice the selected $CC\pi^\pm$ signal of the GENIE central-value prediction used to build the response, consistent with the well-documented generator spread in $CC\pi^\pm$ production on argon [52]. The unfolding correctly recovers the injected NuWro truth (Section C), confirming that the $\sim 2\times$ generator difference is propagated faithfully rather than absorbed into an efficiency or normalisation artefact.

B.5 Software trigger (verified to be a no-op)

A hypothesis that the absence of a `swtrig == 1` filter on MC events was causing a $\sim 13\%$ over-normalisation was investigated. The cut was implemented and the full pipeline rerun; the result was byte-identical. All MC events passing the topological cut already carry `swtrig == 1`: the $\sim 13\%$ of raw MC events failing `swtrig` also fail the topological cut. The cut has been retained for formal correctness but contributes nothing to the current normalisation.

Detector variations. The detector systematic of the pass described here (9.6–14.6%; the current values are 15.6–35.0%, Table 34 of the analysis note) was built from the Run-1 FHC detector variations applied globally. The multi-run FHC, RHC, and combined results instead use *native* Run-4 detector variations per horn mode (the standard eight-knob set), with the combined result accumulating the Run-4 FHC *and* Run-4 RHC detVar samples, each scaled to its own mode’s data exposure, which the framework now supports via multi-file detVar summing. (The Run-5 FHC Recomb2 and SCE detVar productions turned out to be intrinsic- ν_e overlays with zero ν_μ content; the Run-4 FHC knobs stand in for those two.) Likewise, adding the GENIE spline weight to the central-value weight was a verified no-op: `weightSpline` $\equiv 1$ in the NuMI ntuples and the NuWro overlay’s CV weights are sentinel (-1) values clamped to unity.

C NuWro Fake-Data Closure and Cross-Pipeline Validation

Provenance. 2026-07 pass, on the GENIE detVar-CV and NuWro fake data of that era; it predates the weighting fixes, the width-aware regularisation and the released binnings and cannot be regenerated from the repository (analysis note, Limitations). Its qualitative conclusions, no Wiener-SVD bias and NuWro/GENIE of order two, survive; none of its numbers is current.

As a generator-model closure and an independent cross-check of the extraction, the XSE-CANALYZER framework used throughout this note was additionally run (`ProcessNTuples` $\rightarrow$ `UniverseMaker` $\rightarrow$ `Unfolder`) with a NuWro overlay as *fake data* in place of the GENIE detVar-CV sample of the main results. The NuWro reconstructed spectrum is unfolded with the GENIE detector response, efficiency, and GENIE-predicted background subtraction, then compared to the GENIE MicroBooNE-tune prediction. If the unfolding is unbiased, the ratio reflects the genuine NuWro-versus-GENIE model difference rather than a procedural artefact.

Extending the framework beyond p_μ required six additional output branches in the `CC1mu1piXp` selection (reconstructed and true pion momentum, muon $\cos\theta$, and pion $\cos\theta$); the reconstructed pion momentum uses the proton-hypothesis range kinetic energy

(`trk_energy_proton_v`), matching the custom pipeline. The true $\theta_{\mu\pi}$ opening angle, previously written as all-zeros, was also corrected, and all fourteen samples were re-processed.

Table 29: Configuration of the NuWro fake-data closure.

Item	Value
Fake data	NuMI FHC NuWro overlay, 6.6504×10^{20} POT native, high statistics (<code>onBNB</code>)
Exposure	all MC/EXT/dirt/detVar scaled to NuWro POT (high-statistics closure)
Response/eff.	GENIE ν_μ MC overlay (central value)
Unfolding	Wiener-SVD, second-derivative regularisation
Systematics	full multisim + detector-variation universes
Observables	all five (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$)

The NuWro fake data unfolds to approximately twice the GENIE MicroBooNE tune in every observable, uniformly in shape (Fig. 7, Table 30); the two independent unfolding chains recover the same ratio, cross-validating both. The flat $\sim 2\times$ offset here is a genuine NuWro-versus-GENIE-tune $\text{CC}\pi^\pm$ model difference in this NuMI FHC phase space: NuWro is used at its raw cross section (its `weightTune` and `ppfx_cv` branches are unset and default to unity, so no GENIE tune is applied to the fake data). This must *not* be conflated with the apparent $\sim 2\times$ seen in the primary GENIE-detVar-CV result: the two coincide in magnitude but not in cause; the primary-result excess is a POT/flux *bookkeeping* offset from using the detVar-CV sample as fake data ($1.29\times$ more signal events/POT, times the A_C smearing; Sec. B), whereas the NuWro-closure excess is a real generator difference. A caveat on the latter is that part of the excess may be NuWro’s larger *background* being attributed to signal, since the procedure subtracts GENIE-predicted rather than NuWro backgrounds.

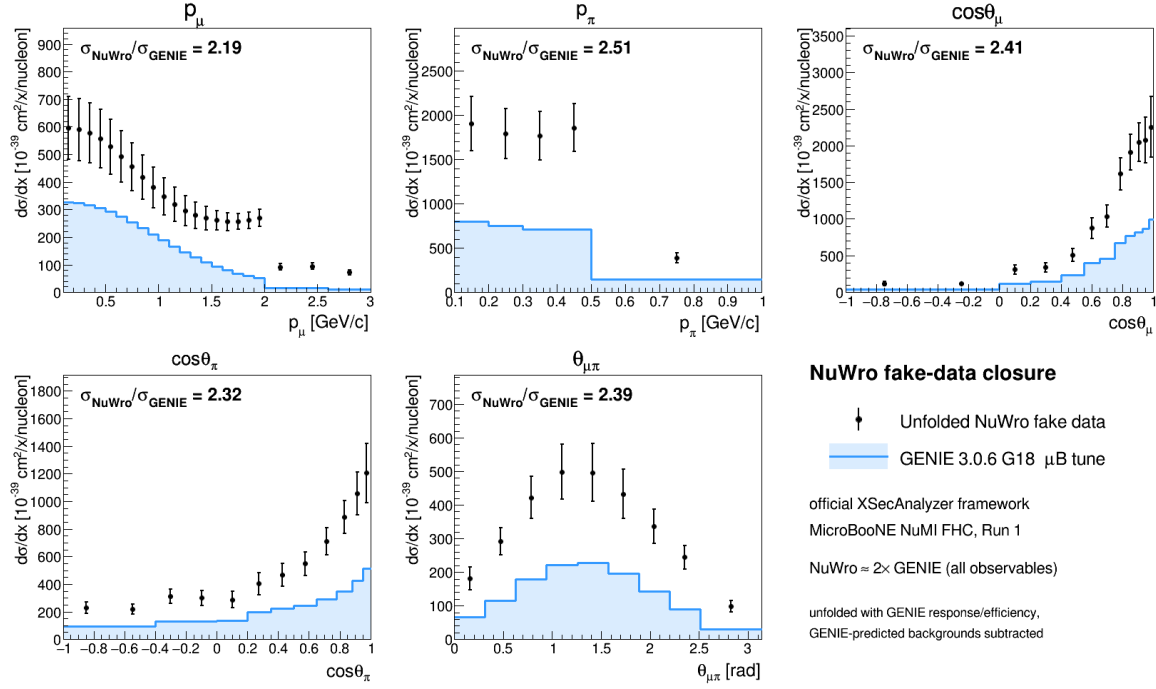


Figure 7: **[HISTORICAL — 2026-07-25, not regenerated.]** This figure predates the central-value weighting fix, the width-aware regularisation and the removal of the duplicated A_C , and it has no producer in the repository, so it cannot be rebuilt from the current state. What survives it is the qualitative point — a NuWro-over- GENIE ratio of order two, which is a genuine generator difference — not the absolute normalisation or the model curve, the latter of which carries the double smearing. NuWro fake-data closure in the official XSECANALYZER framework. Points: unfolded NuWro fake data; shaded line: GENIE 3.0.6 G18 μ B tune. The unfolded NuWro result sits a uniform $\approx 2\times$ above the GENIE prediction in all five observables. **This figure predates the fake-data weighting fix of Sec. 16 of the analysis note:** it was produced 2026-07-25, from a separate cross-pipeline study, and has no producer in the repository. The $\approx 2\times$ NuWro-over-GENIE ratio it shows is a generator comparison and survives the fix; the absolute normalisation of the points does not. Retained as a qualitative cross-pipeline check only.

Table 30: **[HISTORICAL, 2026-07; ratio column not trustworthy.]** NuWro fake-data closure: ratio of integrated cross sections and the χ^2 of the unfolded NuWro result relative to the GENIE tune. The ratio column was found on 2026-09-06 to be identical to the identity-regularised self-closure column of Table 31, so it does not show the NuWro/GENIE ratio of order two that the text and Fig. 7 describe; the original values are not recoverable from the repository. The $p_\mu \chi^2$ is undefined because its 22-bin Wiener-SVD covariance is numerically singular (condition number $\sim 4 \times 10^{13}$).

Observable	$\sigma_{\text{NuWro}}/\sigma_{\text{Genie}}$	χ^2/bins
$d\sigma/dp_\mu$	1.137	n/a
$d\sigma/dp_\pi$	1.125	20.3/5
$d\sigma/d\cos\theta_\mu$	1.159	35.0/12
$d\sigma/d\cos\theta_\pi$	0.931	34.0/12
$d\sigma/d\theta_{\mu\pi}$	1.262	25.8/9

Technical notes. Two framework-level issues were resolved. First, the selection’s opening-angle cut ($\theta_{\mu\pi} < 2.6$ rad) leaves the reconstructed bin $[2.827, \pi)$ structurally empty, producing a zero

covariance diagonal and a Wiener-SVD inversion failure; this was cured by merging the last two $\theta_{\mu\pi}$ bins into $[2.513, \pi)$, giving nine bins. Second, the χ^2 computation in the diagnostic plotter was made robust to the singular p_μ covariance (it now falls back to a sentinel rather than aborting), so the closure plots render for all observables.

C.1 Extraction-bias investigation and algorithm equivalence

Provenance. Same 2026-07 pass as §C; superseded in its numbers (the current closure mean is 1.010, analysis note).

The fake-data closures of that pass showed a consistent $\sim 1.3\times$ ratio of the unfolded total to the raw generator truth, and the detector-variation central-value fake data sits $\sim 1.29\times$ above the GENIE tune. Because a persistent multiplicative offset in a closure test can indicate an unfolding bug, this was investigated end-to-end; the conclusion is that **there is no framework Wiener-SVD bias**.

Algorithm equivalence. The framework unfolders (`WienerSVDUnfolder.cxx`) was compared line-by-line with the independent `RunWienerSVD.FW` implementation used for the custom pipeline. The two are algorithmically identical: covariance whitening $Q = \text{chol}(V^{-1})$, response pre-scaling $R = QA$, the SVD of RC^{-1} , the per-mode Wiener filter $W_t = \eta_t/(\eta_t + 1)$ with $\eta_t = (D_C V_C^T C x_{\text{prior}})_t^2$, and the additional smearing matrix $A_C = C^{-1} V_C W_C V_C^T C$. Feeding the framework’s *exact* extracted matrices (smearceptance, prior, data, and full covariance) into a standalone replica reproduces the framework’s unfolded result bin-for-bin ($\sum \hat{x} = 5836.5$ for $\cos\theta_\mu$), and the framework smearceptance column sums equal the raw efficiency.

Self-closure. The decisive test rebuilds the universes with the GENIE ν_μ overlay itself in the `onBNB` slot, so data and response come from the same sample and the detector-variation POT offset is absent by construction. The unfolded total then recovers the truth for all five observables and all three regularisations (Table 31): $\hat{x}/x_{\text{true}} = 0.94\text{--}1.06$. The closure holds even where the A_C diagonal is as low as 0.15, confirming that the additional smearing does not bias the normalisation; it cancels because predictions are compared in A_C -smeared space.

Table 31: GENIE-overlay self-closure: $\hat{x}/x_{\text{true}}$ for the integrated cross section, by observable and regularisation. Data and response are the same sample, removing the fake-data POT offset.

Observable	Wiener-SVD (id.)	Wiener-SVD (2nd-deriv)	D’Agostini (4 it.)
$d\sigma/dp_\mu$	1.137	0.937	1.040
$d\sigma/d\cos\theta_\mu$	1.159	0.973	0.989
$d\sigma/d\cos\theta_\pi$	0.931	0.996	0.992
$d\sigma/dp_\pi$	1.125	1.063	1.059
$d\sigma/d\theta_{\mu\pi}$	1.262	0.996	0.991

Decomposition. The $\sim 1.29\times$ detector-variation offset is a pure POT/flux bookkeeping factor from using the Run-3b detVar central-value sample as fake data (measured flat across all event types, so not a physics or generator difference); the A_C smearing cancels in the closure, and the $\sim 2\times$ NuWro/GENIE ratio (Table 30) is the genuine model difference. None of the three is an unfolding artefact. The closure χ^2 (small, high p -value) is not the most stringent test given the systematics-dominated covariance; the central $\hat{x}/x_{\text{true}}$ ratio is the more discriminating metric and is what establishes the result above. Since the extraction machinery (`ProcessNTuples`→`UniverseMaker`→`Unfolder`) is identical across horn modes, this closure validates the unbiasedness of the FHC, RHC, and combined full-exposure extractions equally.

References

- [1] MicroBooNE Collaboration, “Measurement of the flux-averaged differential ν_e CC1 π cross section in the NuMI beam,” Internal Note v3.2 (2025).
- [2] K. Mistry, “First Measurement of the Flux-Averaged Differential Charged-Current Electron-Neutrino and Antineutrino Cross Section on Argon with the MicroBooNE Detector,” PhD thesis (2021).
- [3] NOvA Collaboration, M. A. Acero *et al.*, “New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment,” *Phys. Rev. D* **98**, 032012 (2018).
- [4] T2K Collaboration, K. Abe *et al.*, “Constraint on the matter–antimatter symmetry-violating phase in neutrino oscillations,” *Nature* **580**, 339–344 (2020).
- [5] DUNE Collaboration, B. Abi *et al.*, “Prospects for Beyond the Standard Model Physics Searches at the Deep Underground Neutrino Experiment,” *Eur. Phys. J. ST* **231**, 3159–3260 (2022).
- [6] J. S. Marshall and M. A. Thomson, “The Pandora Software Development Kit for Pattern Recognition,” *Eur. Phys. J. C* **75**, 439 (2015).
- [7] W. Tang, X. Li, X. Qian, H. Wei, and C. Zhang, “Data Unfolding with Wiener-SVD Method,” *JINST* **12**, P10002 (2017), arXiv:1705.03568 [physics.data-an].
- [8] L. Aliaga *et al.* (MINERvA Collaboration), “Neutrino Flux Predictions for the NuMI Beam,” *Phys. Rev. D* **94**, 092005 (2016).
- [9] C. Andreopoulos *et al.*, “The GENIE Neutrino Monte Carlo Generator,” *Nucl. Instrum. Methods A* **614**, 87–104 (2010).
- [10] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of space charge effects in the MicroBooNE LArTPC using cosmic muons,” *JINST* **15**, P12037 (2020).
- [11] MicroBooNE Collaboration, C. Adams *et al.*, “A Method to Determine the Electric Field of Liquid Argon Time Projection Chambers Using a UV Laser System and Its Application in MicroBooNE,” *JINST* **15**, P07010 (2020).
- [12] MicroBooNE Collaboration, R. Acciarri *et al.*, “The Pandora multi-algorithm approach to automated pattern recognition of cosmic-ray muon and neutrino events in the MicroBooNE detector,” *Eur. Phys. J. C* **78**, 82 (2018).
- [13] MicroBooNE Collaboration, P. Abratenko *et al.*, “Calorimetric classification of track-like signatures in liquid argon TPCs using MicroBooNE data,” *JHEP* **12**, 153 (2021).
- [14] MicroBooNE Collaboration, P. Abratenko *et al.*, “Determination of muon momentum in the MicroBooNE LArTPC using an improved model of multiple Coulomb scattering,” *JINST* **12**, P10010 (2017).
- [15] MicroBooNE Collaboration, P. Abratenko *et al.*, “Novel approach for evaluating detector-related uncertainties in a LArTPC using MicroBooNE data,” *Eur. Phys. J. C* **82**, 454 (2022).
- [16] MicroBooNE Collaboration, P. Abratenko *et al.*, “New CC0 π GENIE model tune for MicroBooNE,” *Phys. Rev. D* **105**, 072001 (2022), arXiv:2110.14028 [hep-ex].

- [17] MicroBooNE Collaboration, P. Abratenko *et al.*, “First double-differential measurement of kinematic imbalance in neutrino interactions with the MicroBooNE detector,” *Phys. Rev. Lett.* **131**, 101802 (2023), arXiv:2301.03700 [hep-ex].
- [18] S. Gardiner *et al.*, “XSecAnalyzer: a cross-section analysis framework for MicroBooNE,” https://github.com/uboone/xsec_analyzer (accessed 2026).
- [19] P. Adamson *et al.*, “The NuMI Neutrino Beam,” *Nucl. Instrum. Methods A* **806**, 279–306 (2016).
- [20] L. Alvarez-Ruso *et al.* (GENIE Collaboration), “Recent highlights from GENIE v3,” *Eur. Phys. J. ST* **230**, 4449–4467 (2021).
- [21] T. Golan, J. T. Sobczyk, and J. Żmuda, “NuWro: the Wrocław Monte Carlo generator of neutrino interactions,” *Nucl. Phys. B Proc. Suppl.* **229–232**, 499 (2012).
- [22] O. Buss *et al.*, “Transport-theoretical description of nuclear reactions,” *Phys. Rept.* **512**, 1–124 (2012); U. Mosel, “Neutrino event generators: foundation, status and future,” *J. Phys. G* **46**, 113001 (2019).
- [23] Y. Hayato and L. Pickering, “The NEUT neutrino interaction simulation program library,” *Eur. Phys. J. ST* **230**, 4469–4481 (2021).
- [24] S. Agostinelli *et al.* (GEANT4 Collaboration), “GEANT4: a simulation toolkit,” *Nucl. Instrum. Methods A* **506**, 250–303 (2003).
- [25] J. Calcutt, C. Thorpe, K. Mahn, and L. Fields, “Geant4Reweight: a framework for evaluating and propagating hadronic interaction uncertainties in Geant4,” *JINST* **16**, P08042 (2021).
- [26] A. Hoecker *et al.*, “TMVA: Toolkit for Multivariate Data Analysis,” PoS **ACAT**, 040 (2007), arXiv:physics/0703039.
- [27] G. D’Agostini, “A multidimensional unfolding method based on Bayes’ theorem,” *Nucl. Instrum. Methods A* **362**, 487–498 (1995).
- [28] X.-G. Lu, L. Pickering, S. Dolan, G. Barr, D. Coplowe, Y. Uchida, D. Wark, M. O. Wascko, A. Weber, and T. Yuan, “Measurement of nuclear effects in neutrino interactions with minimal dependence on neutrino energy,” *Phys. Rev. C* **94**, 015503 (2016).
- [29] A. P. Furmanski and J. T. Sobczyk, “Neutrino energy reconstruction from one-muon and one-proton events,” *Phys. Rev. C* **95**, 065501 (2017).
- [30] D. Rein and L. M. Sehgal, “Coherent π^0 production in neutrino reactions,” *Nucl. Phys. B* **223**, 29–44 (1983).
- [31] C. Berger and L. M. Sehgal, “Partially conserved axial vector current and coherent pion production by low energy neutrinos,” *Phys. Rev. D* **79**, 053003 (2009).
- [32] MicroBooNE Collaboration, P. Abratenko *et al.*, “First Measurement of Energy-Dependent Inclusive Muon-Neutrino Charged-Current Cross Sections on Argon with the MicroBooNE Detector,” *Phys. Rev. Lett.* **128**, 151801 (2022), arXiv:2110.14023 [hep-ex].
- [33] MicroBooNE Collaboration, R. Acciarri *et al.*, “Design and Construction of the MicroBooNE Detector,” *JINST* **12**, P02017 (2017).
- [34] MicroBooNE Collaboration, C. Adams *et al.*, “Design and construction of the MicroBooNE Cosmic Ray Tagger system,” *JINST* **14**, P04004 (2019).

- [35] D. Rein and L. M. Sehgal, “Neutrino Excitation of Baryon Resonances and Single Pion Production,” *Ann. Phys.* **133**, 79–153 (1981).
- [36] K. S. Kuzmin, V. V. Lyubushkin, and V. A. Naumov, “Lepton polarization in neutrino nucleon interactions,” *Mod. Phys. Lett. A* **19**, 2815–2829 (2004).
- [37] K. M. Graczyk and J. T. Sobczyk, “Form factors in the quark resonance model,” *Phys. Rev. D* **77**, 053001 (2008) [Erratum: *Phys. Rev. D* **79**, 079903 (2009)].
- [38] C. Berger and L. Sehgal, “Lepton mass effects in single pion production by neutrinos,” *Phys. Rev. D* **76**, 113004 (2007).
- [39] J. A. Nowak (MiniBooNE Collaboration), “Four-momentum transfer discrepancy in the charged current π^+ production in the MiniBooNE: Data vs. theory,” *AIP Conf. Proc.* **1189**, 243–248 (2009).
- [40] Y. Hayato, “A neutrino interaction simulation program library NEUT,” *Acta Phys. Polon. B* **40**, 2477–2489 (2009).
- [41] G. M. Radecky *et al.* (ANL Collaboration), “Study of single pion production by weak charged currents in low-energy νd interactions,” *Phys. Rev. D* **25**, 1161–1173 (1982).
- [42] T. Kitagaki *et al.* (BNL Collaboration), “Charged-current exclusive pion production in neutrino–deuterium interactions,” *Phys. Rev. D* **34**, 2554–2565 (1986).
- [43] A. Rodriguez *et al.* (K2K Collaboration), “Measurement of single charged pion production in the charged-current interactions of neutrinos in a 1.3 GeV wide band beam,” *Phys. Rev. D* **78**, 032003 (2008).
- [44] A. A. Aguilar-Arevalo *et al.* (MiniBooNE Collaboration), “Measurement of ν_μ -induced charged-current single positive pion production cross sections on mineral oil at $E_\nu \in 0.5$ –2.0 GeV,” *Phys. Rev. D* **83**, 052007 (2011).
- [45] X.-G. Lu *et al.* (MINERvA Collaboration), “Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at $\langle E_\nu \rangle = 3$ GeV,” *Phys. Rev. Lett.* **121**, 022504 (2018).
- [46] B. Eberly *et al.* (MINERvA Collaboration), “Charged pion production in ν_μ interactions on hydrocarbon at $\langle E_\nu \rangle = 4.0$ GeV,” *Phys. Rev. D* **92**, 092008 (2015).
- [47] C. L. McGivern *et al.* (MINERvA Collaboration), “Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon in the few-GeV region using MINERvA,” *Phys. Rev. D* **94**, 052005 (2016).
- [48] MINERvA Collaboration, J. Wolcott *et al.*, “Evidence for Quasielastic Charge Current Antineutrino Scattering Off Protons in Hydrocarbon at $\langle E_{\bar{\nu}} \rangle = 3.5$ GeV,” *Phys. Rev. Lett.* **116**, 081802 (2016).
- [49] K. Abe *et al.* (T2K Collaboration), “Measurement of the muon neutrino charged-current single π^+ production on hydrocarbon using the T2K off-axis near detector ND280,” *Phys. Rev. D* **101**, 012007 (2020), arXiv:1909.03936 [hep-ex].
- [50] R. Acciarri *et al.* (ArgoNEUT Collaboration), “First measurement of the cross section for ν_μ and $\bar{\nu}_\mu$ induced single charged pion production on argon using ArgoNEUT,” *Phys. Rev. D* **98**, 052002 (2018), arXiv:1804.10294 [hep-ex].

- [51] MicroBooNE Collaboration, P. Abratenko *et al.*, “First measurement of ν_e and $\bar{\nu}_e$ charged-current single charged-pion production differential cross sections on argon using the MicroBooNE detector,” *Phys. Rev. Lett.* **135**, 061802 (2025), arXiv:2503.23384 [hep-ex].
- [52] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of single charged pion production in charged-current ν_μ -Ar interactions with the MicroBooNE detector,” *Phys. Rev. D* **113**, 032007 (2026), arXiv:2509.03628 [hep-ex].